

★ WEEK 15 · PART 1 of 3

Bayesian State-Space: Why Point Estimates Lie

Honest Uncertainty for Dynamic Financial Models

Two layers of doubt · one joint posterior · a line versus a fan

We point the MCMC engine of Week 14 at the dynamic models of Weeks 8-10. A filtered point estimate hides its own uncertainty; a Bayesian state-space model gives a full posterior over hidden states and parameters alike.

1

PART 1 · WHY POINT ESTIMATES LIE

From a Confident Line to an Honest Fan

The quant meltdown that punished false confidence, the two layers of uncertainty in a dynamic model, the Bayesian state-space model that captures both, and the joint posterior over states and parameters.

The Quant Quake of August 2007 — Part 1

When confident point estimates met an uncertain market

In early August 2007, many quantitative market-neutral funds suffered sudden, severe, simultaneous losses. Their models — fit to confident point estimates — had not accounted for how crowded and fragile their shared positions really were.

What happened

- Quant long-short funds lost heavily over a few days.
- A rapid deleveraging cascaded through crowded trades.
- Then a partial, whipsaw rebound followed.

The uncomfortable truth

- The models looked precise and well-fit.
- They had no sense of their own fragility.
- Confidence, not the strategy, was the flaw.

The Quant Quake of August 2007 — Part 2

The lesson: model what you don't know

The episode became a lasting lesson: a model that reports one confident number, with no error bars, invites disaster. Honest quantitative work must quantify what it does NOT know — and let that uncertainty shape every decision.

The lesson

- A single estimate hides the range of what's possible.
- Unmodeled uncertainty doesn't vanish — it bites later.
- Quantify doubt; don't paper over it.

How this week responds

- Put full posteriors on dynamic models.
- Report credible bands, not lone lines.
- Size positions to real, stated uncertainty.

Bringing Two Threads Together

Where the whole course converges

1

We have dynamic models

Kalman and state-space models (Weeks 8-9) and HMMs (Week 10) track hidden states that evolve over time.

2

And we have Bayesian computation

Bayesian inference (Week 13) and MCMC (Week 14) give full posteriors instead of single point estimates.

3

This week joins them

Point the MCMC engine at the dynamic models, and get one joint posterior over hidden states AND parameters — with honest bands on everything.

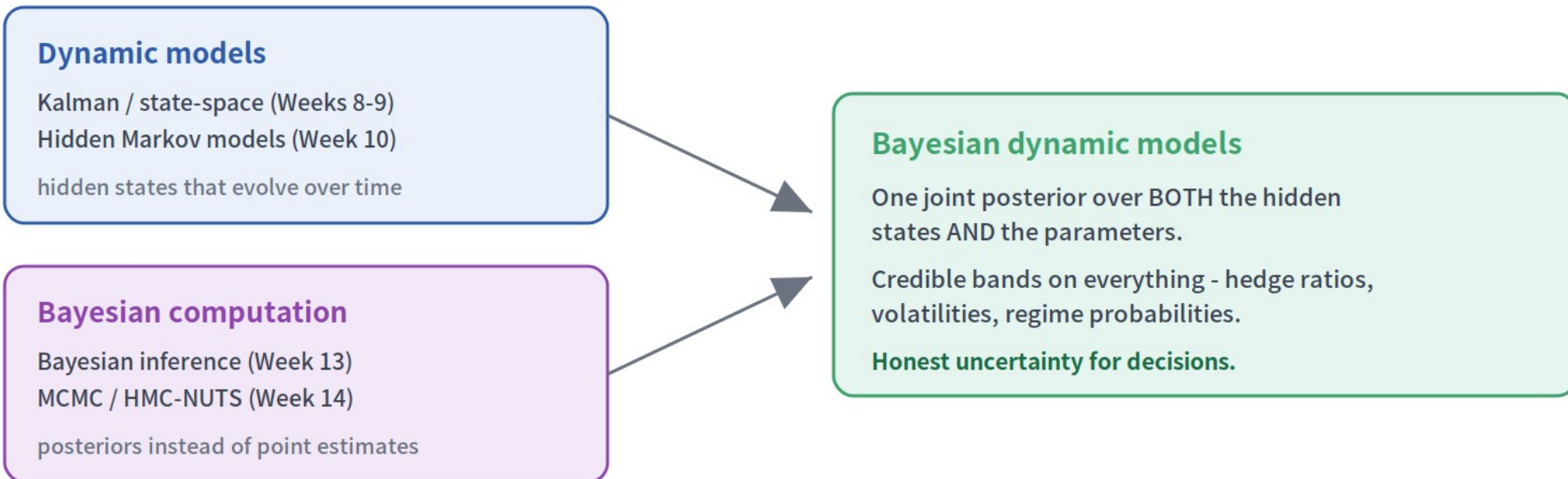
Part 1 in one line: a filtered line hides its uncertainty; a Bayesian state-space model reveals it.

Two Threads of the Course, Joined

The dynamic models of Weeks 8-10 supply hidden states; the Bayesian tools of Weeks 13-14 supply posteriors. Together they give Bayesian dynamic models — one joint posterior, with credible bands on every quantity.

Two Threads of the Course, Joined

Point the Bayesian engine at our dynamic models - and get honest uncertainty over hidden states.



The Gap in Point-Estimate Fitting

Why an MLE-filtered line can lie to you

Fitting a dynamic model by maximum likelihood and then filtering gives a single, smooth line for the hidden state. It looks authoritative — but it fixes the parameters at one estimate and ignores that they, too, are uncertain.

What the line does

- It tracks the hidden state through time.
- It looks clean, precise, and confident.
- It is a single best guess at each moment.

What the line hides

- The parameters were pinned at one estimate.
- Their uncertainty is nowhere in the picture.
- So the line looks far surer than it should.

Why a Filtered Line Looks Too Confident

Fixed parameters, false precision

The hidden assumption

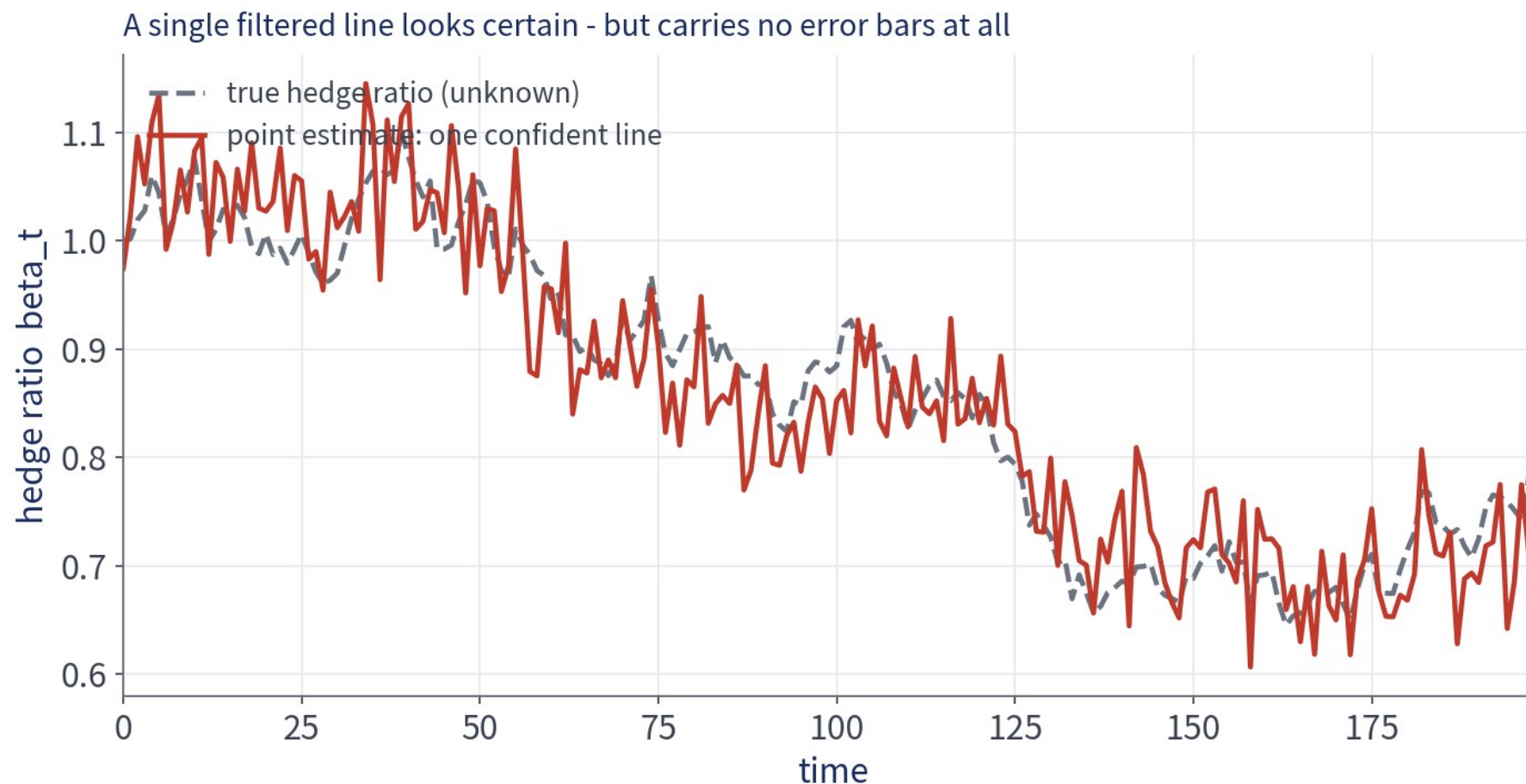
- The filter treats the estimated parameters as exact.
- It reports uncertainty about the state given those values.
- But those values were themselves just estimates.

The consequence

- The reported band is too narrow.
- Decisions built on it are overconfident.
- It is exactly the trap the quant quake exposed.

The Overconfident Point Estimate

A single filtered line for a time-varying hedge ratio looks authoritative and tracks the truth loosely — but it carries no error bars at all. Nothing on the chart tells you how much to distrust it.



Two Layers of Uncertainty

Where doubt comes from in a dynamic model

Honest uncertainty in a dynamic model has two layers. The first is state noise — the hidden state itself moves randomly. The second is parameter uncertainty — we don't know the model's parameters exactly. A truthful band must include both.

Layer 1 — state noise

- The hidden state evolves with random shocks.
- Even known parameters leave the state uncertain.
- This is what a filter already reports.

Layer 2 — parameter uncertainty

- The parameters are estimated, not known.
- Different plausible parameters give different states.
- A point-estimate filter ignores this layer entirely.

State (Process) Noise

The hidden state wanders on its own

What it is

- The state equation includes a random shock each step.
- So the hidden quantity drifts unpredictably.
- The hedge ratio, the volatility, the regime — all move.

How it shows up

- Even with perfect parameters, the state is uncertain.
- The filter's band captures exactly this.
- It is real, but it is only half the story.

Parameter Uncertainty

We don't know the model's dials

What it is

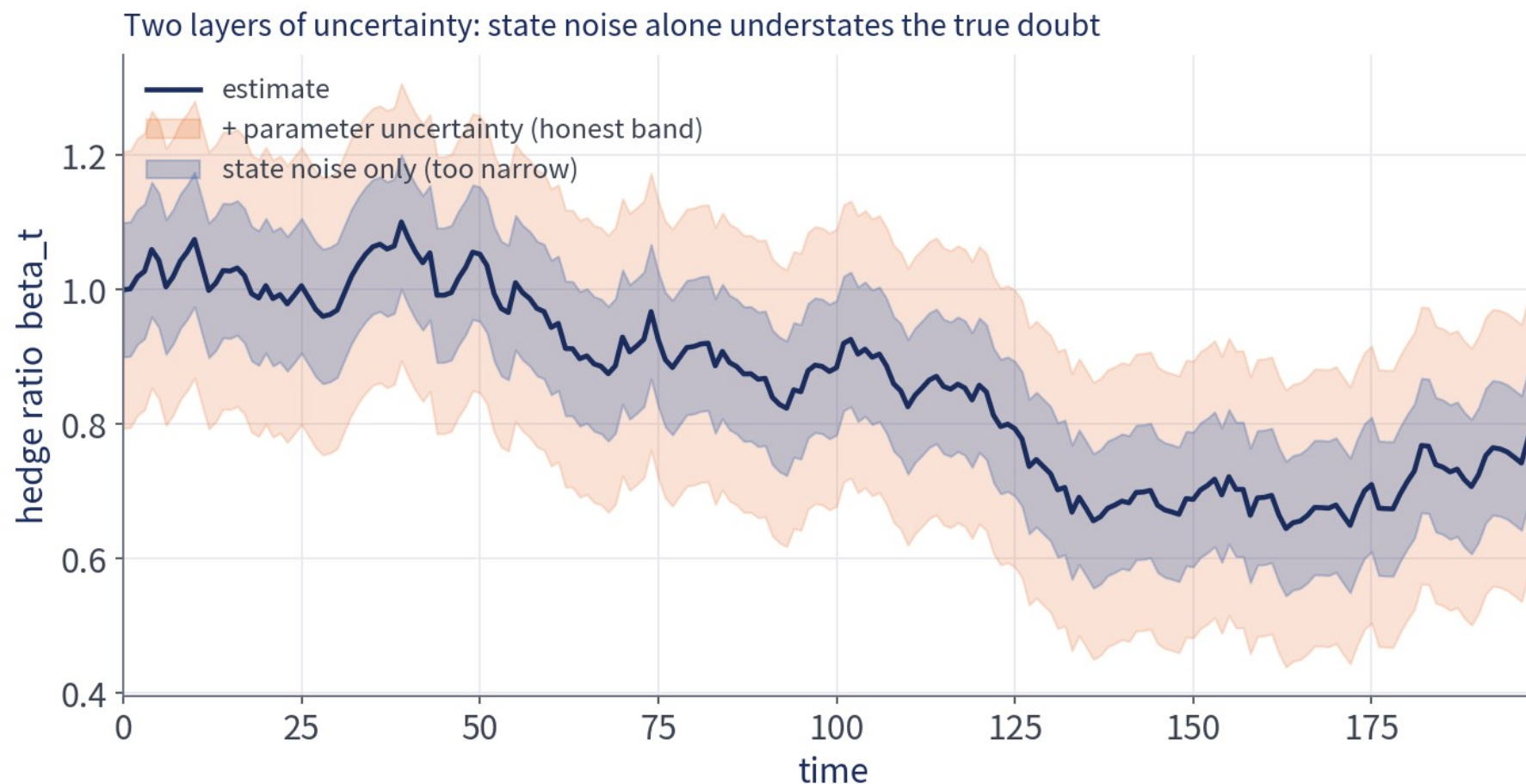
- The noise variances and coefficients are estimated.
- The data pin them down only approximately.
- Several parameter sets fit almost equally well.

Why it matters

- Each plausible parameter set implies a different state path.
- Ignoring this makes the band far too tight.
- The Bayesian approach carries it automatically.

The Two Layers, Side by Side

The inner band is state noise alone — what a filter reports. The wider band adds parameter uncertainty. The honest interval is the wide one; the narrow one quietly overstates how much we know.



The Honest Band Combines Both

One rule adds the two layers

$$\underbrace{\text{Var}(\beta_t \mid y)}_{\text{honest doubt}} = \underbrace{\mathbb{E}[\text{Var}(\beta_t \mid \theta, y)]}_{\text{state noise}} + \underbrace{\text{Var}(\mathbb{E}[\beta_t \mid \theta, y])}_{\text{parameter uncertainty}}$$

Reading it

- Total uncertainty splits into two pieces.
- The average state noise, plus the spread from parameters.
- This is the law of total variance.

The takeaway

- Drop the second term and the band is too narrow.
- A full posterior keeps both by construction.
- That is what makes the band honest.

The Bayesian State-Space Model

One joint posterior over states and parameters

The Bayesian state-space model treats the whole hidden state path AND the parameters as unknowns to infer together. We place priors on the parameters and let the data deliver a single joint posterior over everything at once.

What is unknown

- The entire path of the hidden state.
- The model's parameters — variances and coefficients.
- We infer them jointly, not one then the other.

What we add

- Priors that encode what we know beforehand.
- The same state and observation equations as before.
- The MCMC engine to explore the joint posterior.

The Model in Symbols

A time-varying coefficient, stated cleanly

$$\begin{aligned}\text{observation: } & y_t = \beta_t x_t + \varepsilon_t \\ \text{state: } & \beta_t = \beta_{t-1} + \eta_t\end{aligned}$$

The two equations

- Observation: the data equals a time-varying coefficient times a driver, plus noise.
- State: that coefficient drifts as a random walk.
- This is a dynamic linear model.

What is hidden

- The coefficient path is the hidden state.
- The noise variances are the parameters.
- Priors go on those variances.

The Joint Posterior

States and parameters, inferred together

$$p(\underbrace{\beta_{1:T}}_{\text{states}}, \underbrace{\theta}_{\text{params}} \mid y_{1:T}) \propto p(y_{1:T} \mid \beta_{1:T}, \theta) p(\beta_{1:T} \mid \theta) p(\theta)$$

Reading it

- The posterior is over the whole state path and the parameters.
- It factors into a likelihood, a state prior, and a parameter prior.
- MCMC samples from this joint distribution.

Why joint matters

- States and parameters inform each other.
- Inferring them together propagates both uncertainties.
- That is precisely the honest band we want.

What We Get From It

Credible bands on everything

Not a single answer

- A full posterior over the whole state path.
- A posterior over every parameter.
- Draws you can summarize any way you need.

Bands on real quantities

- A credible band around a time-varying hedge ratio.
- A band around a volatility or a regime probability.
- Uncertainty you can feed into decisions.

Point Estimate vs Full Posterior

One confident line, or an honest fan

Here is the whole shift in one image. The point estimate gives a single line for the hedge ratio. The Bayesian model gives a fan — a spread of plausible paths whose width is the honest uncertainty at every moment.

The line

- One value at each time.
- Clean, but silent about doubt.
- It hides how much it could be wrong.

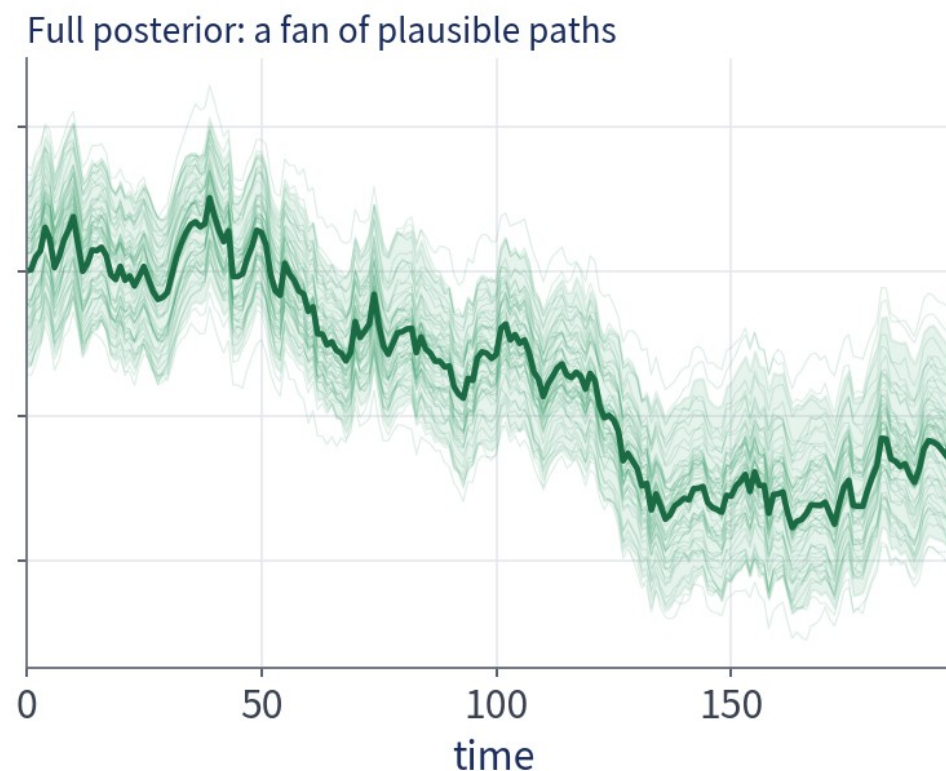
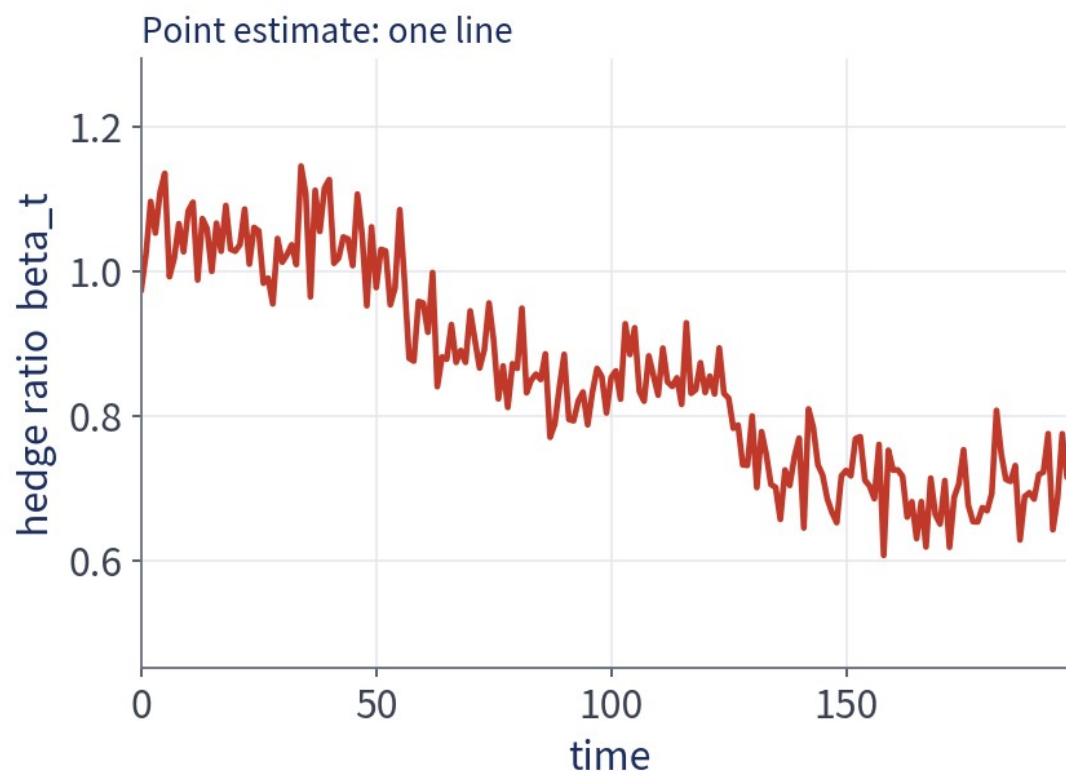
The fan

- A distribution of paths at each time.
- Its width is the uncertainty, stated plainly.
- Narrow where data is rich, wide where it is thin.

The Hedge Ratio: Line vs Fan

On the left, the point estimate is a single confident line. On the right, the posterior is a fan of plausible hedge-ratio paths with a credible band. The fan is the honest object to trade on.

The hedge ratio β_{t-1} : one confident line, or an honest fan of possibilities



Why the Fan Matters for Decisions

Uncertainty should shape the trade

With a line

- You act as if the estimate is exactly right.
- You size the same whether sure or unsure.
- You are blind to the risk of being wrong.

With a fan

- You see when the estimate is shaky.
- You can trade smaller when the band is wide.
- Uncertainty flows through to position size.

What Parts 2 and 3 Add

From the model to the machinery and the switches

Part 2 — sampling and applying

- Sampling states and parameters together.
- Forward-filter, backward-sample for whole paths.
- The Bayesian hedge ratio and stochastic volatility.

Part 3 — switching models

- The Bayesian HMM and soft regime probabilities.
- Structural breaks at uncertain dates.
- Trading with uncertainty — pairs Stage VI.

See the Overconfidence for Yourself

Direct Claude Code to compare a line and a fan

GOAL Fit a time-varying coefficient two ways and see how a point estimate understates the uncertainty.

[INPUT] Prompt · Claude Code

Write and run `line_vs_fan.py` that:

1. simulates a dynamic linear model - a hedge ratio that drifts as a random walk, with noisy observations;
2. fits it with a Kalman filter at point-estimate parameters and plots the single filtered line;
3. also draws many plausible paths accounting for parameter uncertainty, forming a fan with a band;
4. overlays both against the true hidden path.

Show the point band is too narrow and the fan covers the truth honestly.

How to work with it

- The line fixes the parameters.
- The fan carries their uncertainty.
- The fan is wider - and honest.
- It is narrow where data is rich.

Deliverable: `line_vs_fan.py` + the filtered line and the posterior fan against the true path.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

What two layers make up honest uncertainty in a dynamic model?

**State noise +
parameter
uncertainty**

Q2

What does a Bayesian state-space model infer?

**A joint posterior over
states and
parameters**

Q3

What replaces the single confident line?

**A posterior fan with a
credible band**

All three come from the two layers, the joint posterior, and the line-versus-fan contrast.

Three Misconceptions to Avoid

Where state-space intuition slips

Common myths

- "A filter's band already shows all the uncertainty."
- "Point estimates are fine if the model fits well."
- "States and parameters can be fit one then the other."

The reality

- It shows state noise but omits parameter uncertainty.
- A good fit can still be dangerously overconfident.
- Inferring them jointly is what propagates both doubts.

Three Things to Carry Forward

Part 1 in one screen

1

Point estimates hide their uncertainty

A filtered line fixes the parameters at one estimate and reports only state noise. It looks precise but omits parameter uncertainty — the false confidence the quant quake punished.

2

Honest doubt has two layers

State noise and parameter uncertainty together make the true band. The law of total variance adds them; a full posterior keeps both by construction.

3

The Bayesian state-space model gives a fan

It infers one joint posterior over the hidden state path and the parameters, turning a single confident line into an honest fan with credible bands on everything.

Next — Part 2: sampling states and parameters together with FFBS, and the Bayesian hedge ratio.

★ WEEK 15 · PART 1 · PRACTICE

When a Confident Number Costs You

Two Cases, Worked by Hand

A wandering beta and a lucky Sharpe - computed step by step

Part 1 in the real world, with the arithmetic shown. We compute a beta and a Sharpe by hand from tiny datasets, watch each point estimate mislead, and update our belief into an honest band - one line at a time.

1P

PART 1 · APPLICATIONS

Two Cases, Computed by Hand

Case A: compute a stock's beta from five days, see it move, then update it into a band. Case B: compute a Sharpe from six months and find its interval spans zero. Every step is shown; then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

A point estimate hides its uncertainty

A single fitted number looks precise but omits how much it could be wrong - especially the parameter uncertainty a filter ignores.

2

Honest doubt has two layers

The quantity itself moves, and we don't know the model's dials. A truthful band includes both, and we will see it emerge from the arithmetic.

3

A posterior turns a line into a fan

A Bayesian update replaces the confident line with a band - and it is just a weighted average you can compute by hand.

Everything here is done with a calculator: five days of data, six months of returns, one update rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the point estimate by hand, step by step.
- Then find its honest band - also by hand.

What you should watch

- How the single number looks confident.
- How wide the honest band actually is.
- How the band would change the decision.

The Wandering Market Beta

Every hedge and cost-of-capital rests on it

A stock's beta - its sensitivity to the market - sets the hedge you put on and the discount rate you value it with. It is usually reported as one number. We will compute that number from five days of data, then watch it move and learn to put a band on it.

Where beta is used

- Hedging: how much index to short.
- Cost of capital: the discount rate.
- Risk: the stock's market exposure.

What we'll do by hand

- Estimate beta from five days.
- See two windows disagree.
- Update belief into a band.

Beta as a Drifting Hidden State

The dynamic CAPM, in state-space form

$$\begin{aligned}\text{return: } r_{i,t} &= \alpha + \beta_t r_{m,t} + \varepsilon_t \\ \text{drift: } \beta_t &= \beta_{t-1} + \eta_t\end{aligned}$$

The two equations

- Return: stock equals alpha plus beta times the market, plus noise.
- Drift: beta wanders as a random walk.
- Beta is the hidden state we infer.

Why state-space

- It lets beta move, not stay fixed.
- It yields a band, not just a line.
- The Part 1 model, applied to CAPM.

The Data: Five Days

Compute this stock's beta by hand

Day	Market r_m (%)	Stock r_i (%)
1	+1.0	+1.4
2	-0.5	-0.3
3	+2.0	+2.6
4	-1.0	-1.0
5	+0.5	+0.8

Goal: estimate $\beta = \text{Cov}(\text{market}, \text{stock}) / \text{Var}(\text{market})$. We do it in three lines on the next slide.

Beta = Cov / Var

Three lines with a calculator

$$\begin{aligned}\bar{r}_m &= \frac{1.0-0.5+2.0-1.0+0.5}{5} = 0.40, & \bar{r}_i &= \frac{1.4-0.3+2.6-1.0+0.8}{5} = 0.70 \\ \text{Cov} &= \frac{1}{5} \sum (r_m - \bar{r}_m)(r_i - \bar{r}_i) = \frac{6.75}{5} = 1.35 \\ \text{Var} &= \frac{1}{5} \sum (r_m - \bar{r}_m)^2 = \frac{5.70}{5} = 1.14 \\ \hat{\beta} &= \frac{\text{Cov}}{\text{Var}} = \frac{1.35}{1.14} = \mathbf{1.18}\end{aligned}$$

The single best estimate is beta = 1.18. It looks exact - but it came from just five noisy days.

Two Windows, Two Betas

The number moves as the window moves

Slide the window forward

- Days 1-5 gave beta = 1.18.
- The next five days might give, say, 0.90.
- Same stock, a very different number.

What this tells you

- The 'one beta' is really a moving target.
- Each window is a noisy snapshot.
- No single number is the honest answer.

The Rolling-Regression Beta

Repeating that five-day calculation on a sliding window gives one jumpy line: it moves around a lot yet reports no uncertainty. Is a jump real signal, or just window noise? The line cannot say.



Update the Belief Into a Band

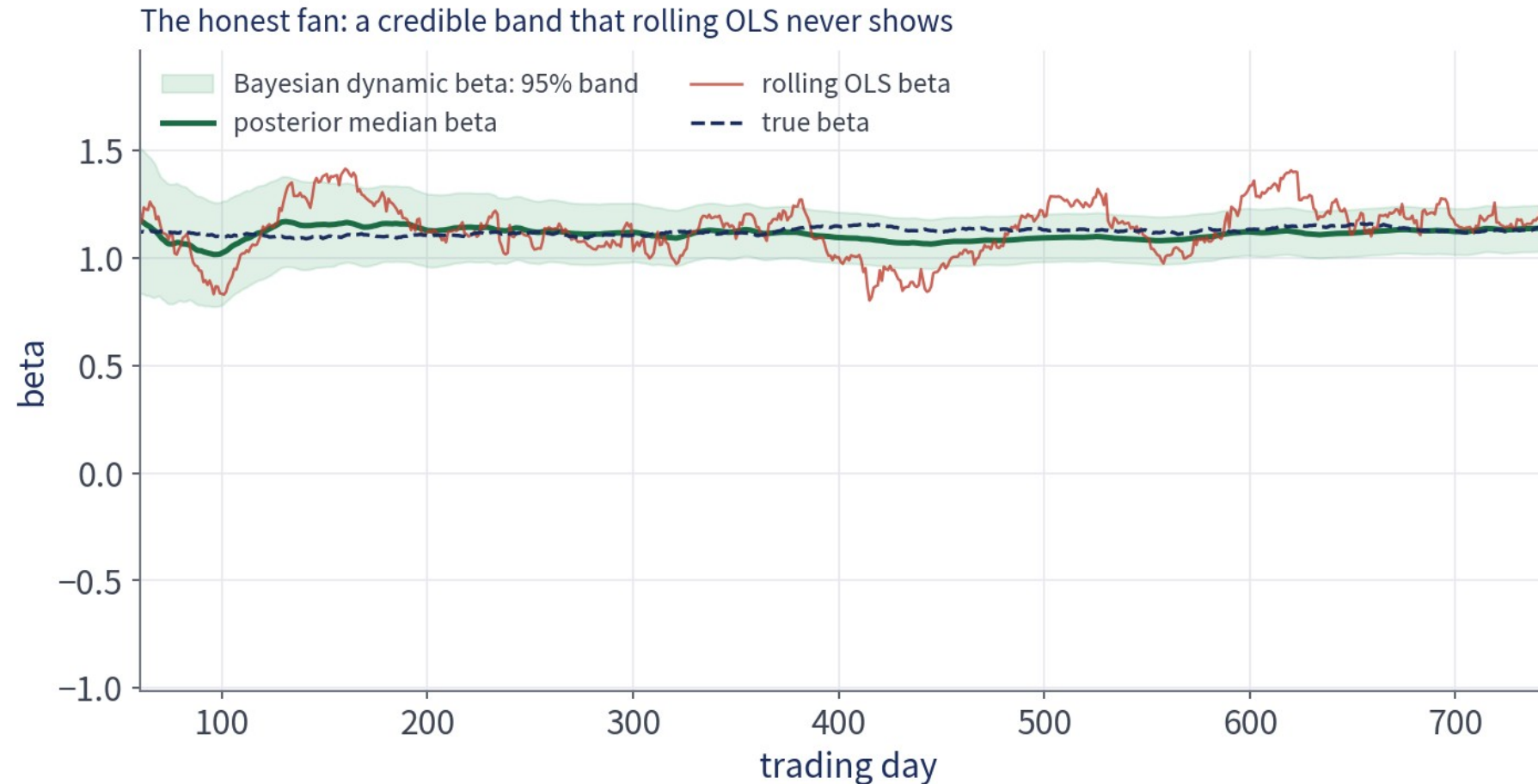
Prior + data = posterior, by hand

$$\begin{aligned}\text{prior } \beta &\sim N(1.0, 0.20^2), & \text{new estimate } y &= 1.4 \pm 0.10 \\ \text{precision} &= \frac{1}{0.20^2} + \frac{1}{0.10^2} = 25 + 100 = 125 \\ \text{posterior var} &= \frac{1}{125} = 0.008 \Rightarrow \text{sd} = \sqrt{0.008} = 0.089 \\ \text{posterior mean} &= \frac{25(1.0) + 100(1.4)}{125} = \frac{165}{125} = \mathbf{1.32} \pm 0.089\end{aligned}$$

The posterior is a precision-weighted average: pulled toward the sharper input, and tighter than both.
That sd is the band.

The Honest Fan for Beta

Repeating that update every day gives the Bayesian dynamic beta: it tracks the true path smoothly AND carries a band. Where the rolling line jumps confidently, the band widens to tell you beta is hard to pin down



What the Overconfident Beta Costs

The band changes real decisions

With the single number

- You hedge as if $\beta = 1.18$ exactly.
- You discount cash flows at one precise rate.
- A wrong beta mis-sizes the hedge and the value.

With the band

- You see when beta is well-pinned or shaky.
- You widen the hedge margin when it is uncertain.
- You stress the valuation across the range.

Skill, or Just Luck?

A great track record - or a short one

A strategy shows a two-year record with a headline Sharpe ratio. Impressive - the kind of number that wins an allocation. We will compute that Sharpe from six months by hand, then compute its error bar, and find the honest interval cannot even rule out zero.

The pitch

- A smooth-looking equity curve.
- A Sharpe that reads as real skill.
- It attracts capital fast.

What we'll do by hand

- Compute the sample Sharpe.
- Compute its standard error.
- Build its 95% interval.

The Track Record: Six Months

Compute the Sharpe by hand

Month	1	2	3	4	5	6
Return (%)	+2.5	-1.0	+3.0	+1.5	-0.5	+3.5

Goal: Sharpe = mean return / standard deviation, then annualize by multiplying by the square root of 12.

The Sample Sharpe

Mean, spread, ratio, annualize

$$\bar{r} = \frac{2.5 - 1.0 + 3.0 + 1.5 - 0.5 + 3.5}{6} = 1.50\%$$

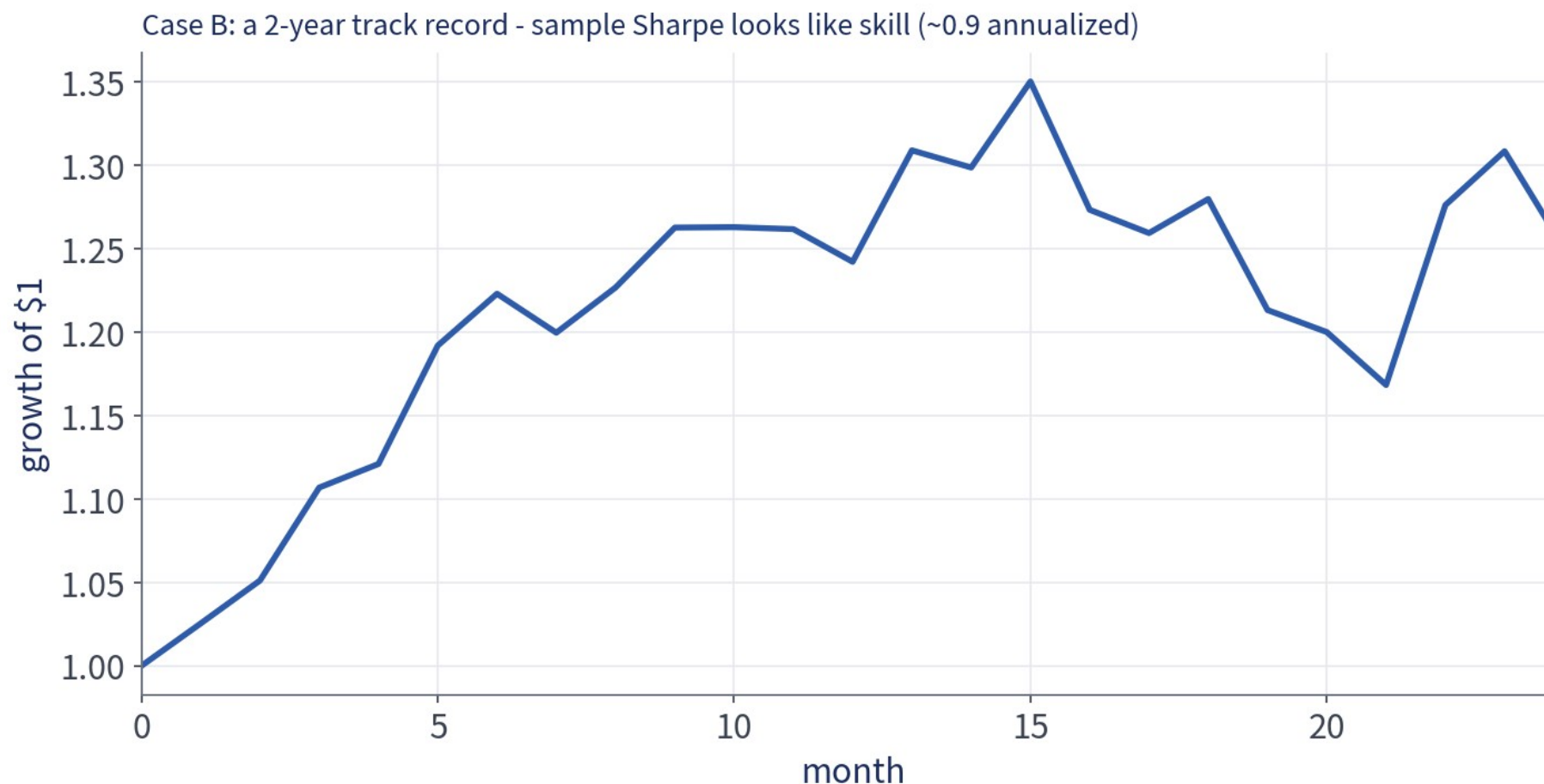
$$s = \sqrt{\frac{\sum (r - \bar{r})^2}{6 - 1}} = \sqrt{\frac{17.5}{5}} = \sqrt{3.5} = 1.87\%$$

$$SR_{\text{month}} = \frac{1.50}{1.87} = 0.80, \quad SR_{\text{ann}} = 0.80 \times \sqrt{12} = \mathbf{2.78}$$

An annualized Sharpe of 2.78 looks spectacular. But it rests on only six data points - so how sure is it?

The Impressive-Looking Track Record

Plotted, those months compound into steady growth with a headline Sharpe. As a point estimate it looks like a clear win - exactly the number that attracts capital. But a point estimate is all it is.



How Uncertain Is a Sharpe?

The formula we'll plug into

$$\widehat{SR} \pm se, \quad se \approx \sqrt{\frac{1 + \frac{1}{2}\widehat{SR}^2}{T}} \quad (\text{short } T \Rightarrow \text{wide band})$$

Reading it

- The Sharpe comes with a standard error.
- It shrinks only with more time, T .
- Six months is very little T .

What to expect

- A wide band around the estimate.
- A headline near three, consistent with zero.
- We compute it exactly next.

The 95% Interval Spans Zero

Plug in and see

$$\text{se}(SR_{\text{month}}) = \sqrt{\frac{1 + \frac{1}{2}SR^2}{T}} = \sqrt{\frac{1 + 0.5(0.80)^2}{6}} = \sqrt{0.220} = 0.469$$

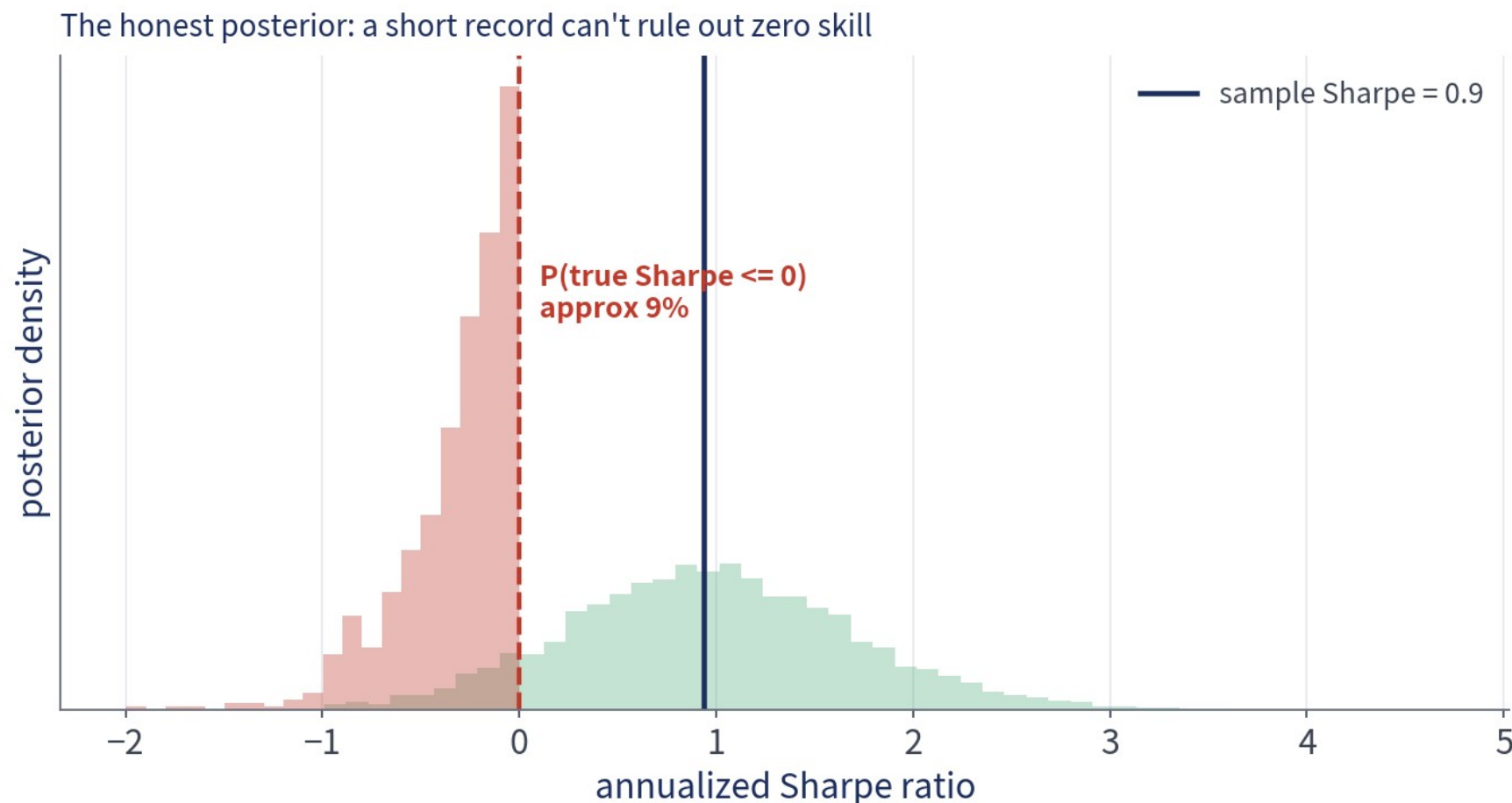
$$\text{se}(SR_{\text{ann}}) = 0.469 \times \sqrt{12} = 1.63$$

$$95\% \text{ CI} = 2.78 \pm 1.96(1.63) = 2.78 \pm 3.19 = (-\mathbf{0.41}, \mathbf{5.96}) \ni 0$$

Even a Sharpe of 2.78 has a 95% interval from -0.41 to 5.96. It includes zero: over six months, skill and luck are indistinguishable.

The Posterior Spans Zero

Resampling the returns gives the same message as the hand calculation: the Sharpe's distribution is wide, and a real share of it sits at or below zero. A confident headline hides a genuine chance of no edge.



What the Overconfident Sharpe Costs

The quant-quake lesson, in miniature

With the single number

- You allocate as if the skill is proven.
- You size the position to a Sharpe near three.
- You may be paying for luck, at full size.

With the interval

- You allocate in proportion to the evidence.
- You start small and add as the record lengthens.
- You demand more data before betting big.

The Common Pattern

Two calculations, one shape

Both cases share a shape you have now computed by hand: a single fitted number drives a real decision while hiding how uncertain it is. Beta sets a hedge; the Sharpe wins an allocation. In each, a Bayesian band turns the deceptive line into an honest fan.

The trap

- A number you act on, taken as exact.
- Uncertainty invisible in the point estimate.
- A decision sized to false precision.

The fix

- Compute the band, not just the point.
- Weight prior and data by precision.
- Size the decision to that honest width.

Where a Confident Number Bites

Beyond beta and the Sharpe, the same trap recurs: factor loadings, volatility and VaR, hedge ratios, duration. Wherever one fitted number drives a decision, the same hand-computable band reveals the risk it hid

Where a Confident Number Bites

The same trap - a number you act on that hides its uncertainty - recurs all over finance.

Market beta

used for: hedging, cost of capital

one line -> a fan

Sharpe / alpha

used for: manager selection

a point -> spans zero

Factor loadings

used for: risk attribution

fixed -> uncertain

VaR / volatility

used for: risk limits

one level -> a band

Hedge ratio

used for: pairs trading

line -> credible band

Duration / DV01

used for: rate hedging

point -> posterior

The pattern, every time

Wherever a single fitted number drives a real decision, a Bayesian state-space model replaces it with a posterior - and the width of that posteri

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full time series.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - beta:

- * take the 5-day table; compute Cov, Var, beta and print each intermediate number;
- * do the normal-normal update by hand-formula, printing posterior mean and sd;
- * then run a rolling beta and a dynamic-beta band.

CASE B - Sharpe:

- * take the 6-month returns; print mean, sd, monthly and annual Sharpe;
- * compute the SE and the 95% interval;
- * confirm it includes zero, then bootstrap it.

Match every printed number to the slides.

Check against the slides

- $\text{beta} = 1.18$; posterior 1.32 ± 0.09 .
- Sharpe (ann.) = 2.78.
- 95% interval $(-0.41, 5.96)$.
- Both include a value that flips the call.

Deliverable: `part1_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get beta from the five-day table?

$\text{Cov}(\text{market}, \text{stock}) / \text{Var}(\text{market})$

Q2 In the update, why is the posterior pulled toward the data?

The data had higher precision

Q3 Why can't a 6-month Sharpe of 2.78 prove skill?

Its 95% interval includes zero

All three come from the beta calc, the precision-weighted update, and the Sharpe interval.

Two Calculations to Remember

Part 1 applications in one screen

1

A beta is Cov/Var - and it moves

We computed $\beta = 1.18$ from five days, saw a second window disagree, then updated prior and data into a posterior of 1.32 ± 0.09 . The band is a precision-weighted average, tighter than either input.

2

A short Sharpe can't tell skill from luck

Six months gave an annualized Sharpe of 2.78 - but its 95% interval runs from -0.41 to 5.96. It includes zero, so the impressive number proves nothing yet.

3

Same trap, same fix

Wherever one hand-computed number drives a decision, compute its band too. The width of that band is the risk the point estimate was hiding.

Next - Part 2 practice: a hedge ratio and a VaR, also worked by hand.

★ WEEK 15 · PART 2 of 3

Sampling States and Parameters Together

FFBS, the Bayesian Hedge Ratio, and Stochastic Volatility

Two-block Gibbs · forward-filter backward-sample · TVP-SV

Now the machinery. A two-block Gibbs sweep alternates between states and parameters; FFBS draws a whole state path at once. We put it to work on a time-varying hedge ratio, then add a breathing volatility.

2

PART 2 · SAMPLING THE MODEL

From Joint Posterior to Draws

The two-block Gibbs sweep, forward-filter backward-sample for whole state paths, the Bayesian time-varying hedge ratio with credible bands, and TVP-SV when the noise level itself moves.

From Model to Machinery

Part 1 defined it; Part 2 samples it

1

We want a joint posterior

Part 1: infer the hidden state path and the parameters together, and get an honest fan instead of a confident line.

2

But sampling it looks daunting

The state path can have hundreds of entries, all correlated, entangled with the parameters. How do we draw from that?

3

Split it into two easy blocks

Draw the states given the parameters, then the parameters given the states. Each half is easy; alternating explores the joint posterior.

Part 2 in one line: alternate states and parameters, draw whole paths with FFBS, apply it to the pair.

Sampling States and Parameters Together

A two-block Gibbs sweep for dynamic models

The joint posterior over a long state path and the parameters is a tangle — but it untangles if you condition. Given the parameters, the states are a clean linear-Gaussian problem; given the states, the parameters are a simple regression. Gibbs alternates between the

two

The hard version

- States and parameters are correlated and coupled.
- Sampling them jointly, all at once, is intractable.
- The state path alone can be hundreds of unknowns.

The easy split

- States given parameters: a Kalman problem.
- Parameters given states: a standard regression.
- Alternate the two, and the tangle dissolves.

The Two-Block Gibbs Sweep

States, then parameters, then repeat

block 1 (states): $\beta_{1:T} \sim p(\beta_{1:T} \mid \theta, y)$ (FFBS)
block 2 (params): $\theta \sim p(\theta \mid \beta_{1:T}, y)$

Block one

- Fix the parameters at their current draw.
- Draw the entire state path in one go (FFBS).
- This is the linear-Gaussian half.

Block two

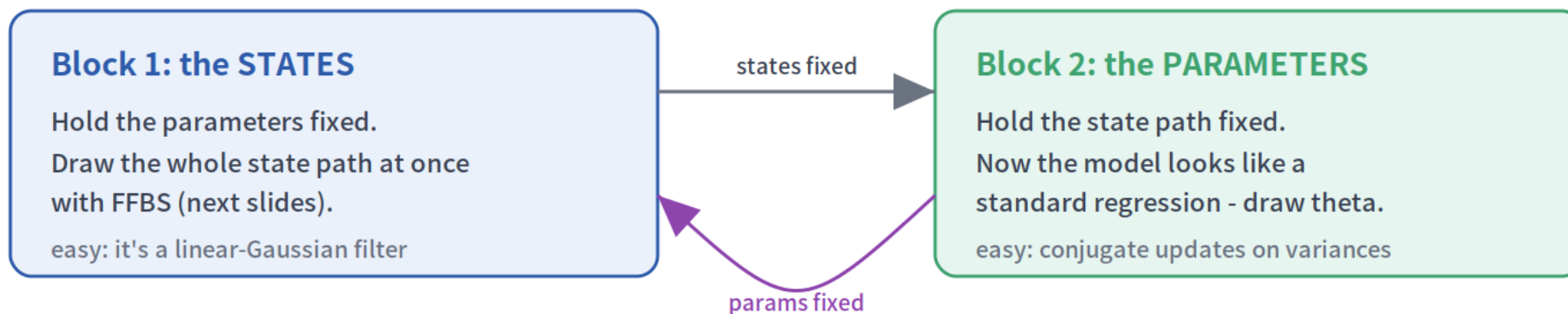
- Fix the state path just drawn.
- Draw the parameters from their conditional.
- For the variances, these are conjugate updates.

A Two-Block Gibbs Sweep

Hold the parameters and draw the whole state path with FFBS; then hold the states and draw the parameters as a regression. Each conditional is easy, and repeating the sweep explores the full joint posterior

A Two-Block Gibbs Sweep

Alternate: draw all the states given the parameters, then the parameters given the states.



Why split it in two

Jointly, states and parameters are a hard tangle. But EACH given the other is easy: states given parameters is a Kalman problem; parameters given states is a simple regression. Alternating explores the joint posterior.

Repeat the sweep thousands of times - the draws are the joint posterior.

Draw the States Given the Parameters

A clean linear-Gaussian problem

What makes it easy

- With parameters fixed, the model is linear-Gaussian.
- That is exactly the Kalman setting from Week 9.
- The whole state path can be drawn at once with FFBS.

What we get

- One complete, consistent draw of the hidden path.
- Not just a mean — an actual sampled trajectory.
- Fed forward to the parameter step.

Draw the Parameters Given the States

Now it's just a regression

What makes it easy

- With the state path fixed, the states are 'known'.
- The model collapses to a standard regression.
- The noise variances get simple conjugate updates.

What we get

- A fresh draw of the parameters.
- Consistent with the current state path.
- Fed back into the next state-sampling step.

Forward-Filter, Backward-Sample

How to draw a whole state path at once

Block one needs a full path, not just filtered means. FFBS does it in two passes: run the Kalman filter forward to gather information, then sweep backward, drawing each state given the one after it — yielding one coherent path per sweep.

Why not just filter

- The filter gives a mean and variance at each time.
- But we need a joint DRAW of the whole path.
- Independent per-time draws would be inconsistent.

What FFBS does

- Forward pass: the familiar Kalman filter.
- Backward pass: sample the path end to start.
- The result is one internally consistent trajectory.

The FFBS Factorization

Draw the end, then work backward

$$p(\beta_{1:T} \mid y, \theta) = p(\beta_T \mid y) \prod_{t=1}^{T-1} p(\beta_t \mid \beta_{t+1}, y_{1:t})$$

Reading it

- Start by drawing the final state from the filter.
- Then draw each earlier state given the next one.
- Each conditional uses the forward-filtered quantities.

Why it works

- The chain structure makes each step Gaussian.
- Conditioning on the next state ties the path together.
- One backward pass yields one full sample.

The Forward Filter Pass

Gather information, left to right

What it does

- Run the Kalman filter from start to finish.
- At each time, form the filtered mean and variance.
- Store them — the backward pass will need them.

What it is

- Exactly the filter from Week 9, unchanged.
- It summarizes all data up to each time.
- It is the 'learn' half of FFBS.

The Backward Sampling Pass

Draw a path, right to left

What it does

- Draw the final state from its filtered distribution.
- Step backward, drawing each state given the next.
- Use the stored forward quantities at each step.

What it gives

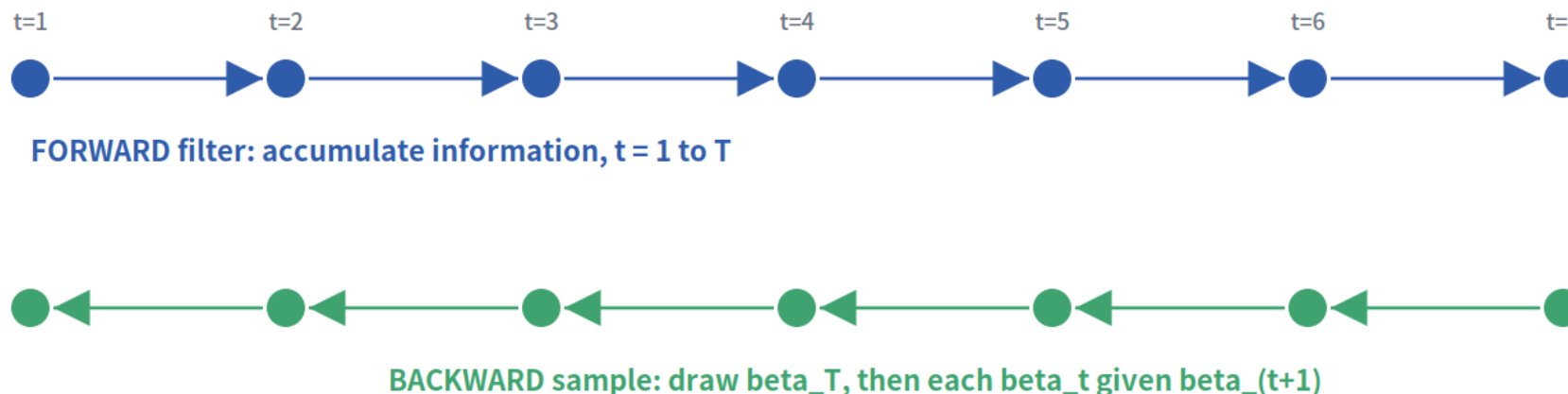
- One complete, coherent state path.
- A genuine sample, not a smoothed average.
- It is the 'draw' half of FFBS.

FFBS: Forward to Learn, Backward to Draw

The forward pass is the Kalman filter, accumulating information from the first observation to the last. The backward pass draws the final state, then each earlier one given its successor — one full path per sweep.

FFBS: Forward to Learn, Backward to Draw

One pass filters left-to-right; a second pass samples a coherent path right-to-left.



The forward pass is the familiar Kalman filter. The backward pass starts at the end and draws each state given the one after it - producing ONE complete, internally consistent path in a single sweep.

Each Sweep Draws One Path

Many sweeps make the fan

One sweep

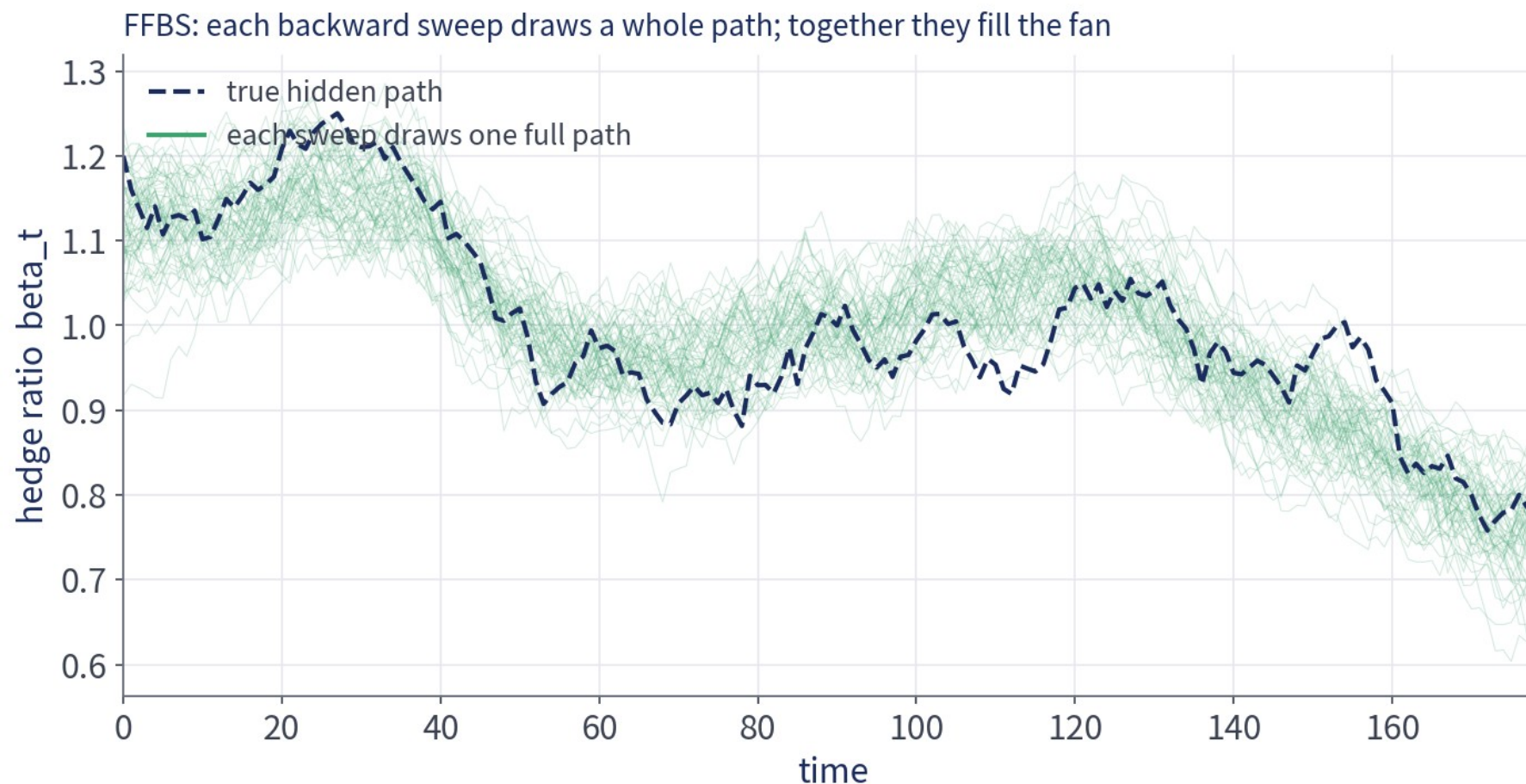
- A single FFBS pass yields one whole path.
- It is a plausible history of the hidden state.
- Different sweeps give different paths.

Many sweeps

- Collect hundreds of sampled paths.
- Together they trace out the posterior fan.
- Their spread at each time is the credible band.

FFBS Builds the Fan

Each backward sweep draws one complete hedge-ratio path. Overlaid, hundreds of these draws fill out the posterior fan around the true hidden path — the raw material for a credible band.



The Bayesian Time-Varying Hedge Ratio

Pairs trading Stage III, now with credible bands

Recall the pairs-trading hedge ratio from Stage III — the coefficient linking the two assets, which drifts over time. Modeled as a Bayesian state-space, it becomes a posterior fan: a median hedge ratio with an honest band at every date.

The Stage III object

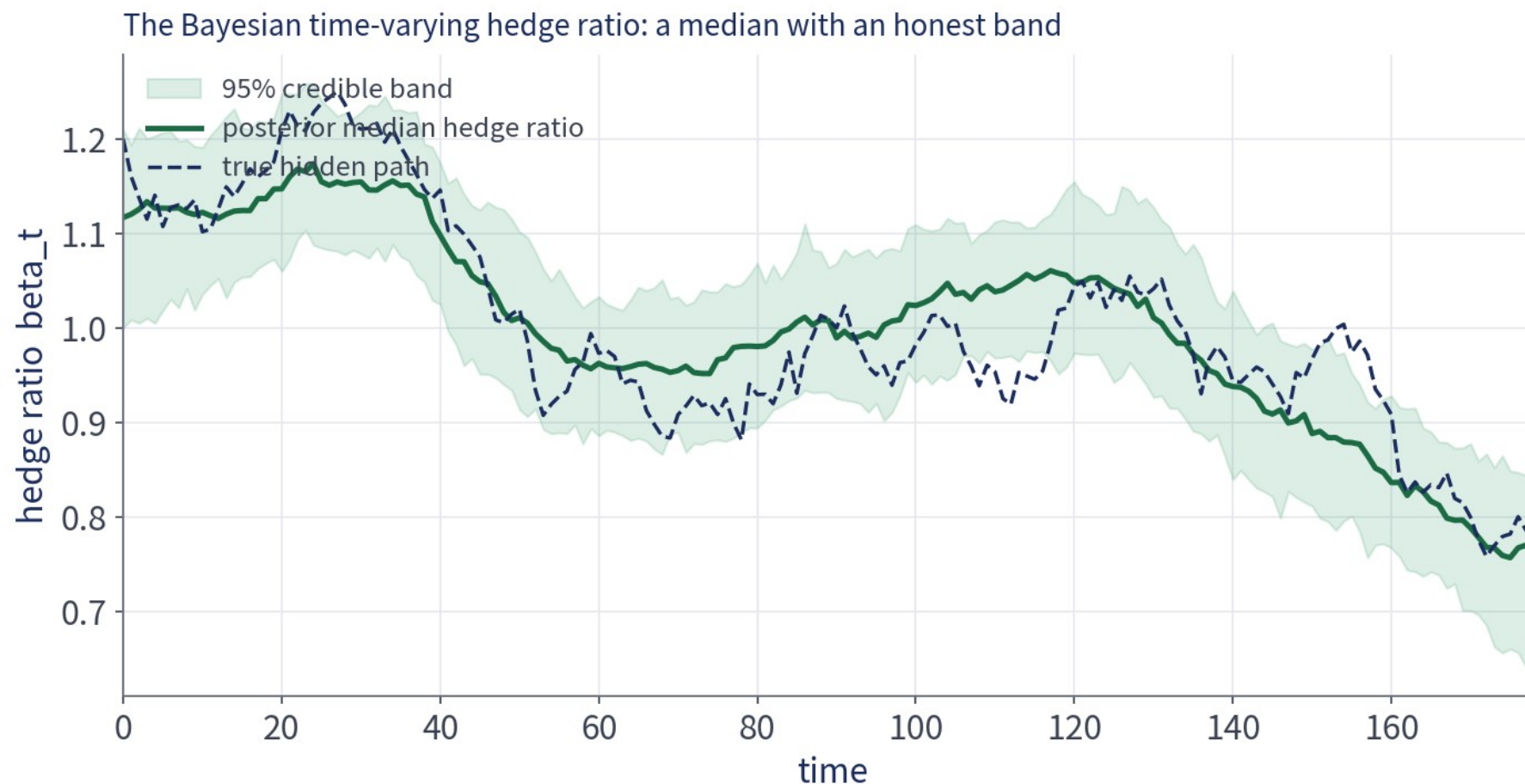
- The hedge ratio relating the two legs of the pair.
- It is not constant — it drifts as markets change.
- Earlier we tracked it with a single filtered line.

The Bayesian upgrade

- Now a posterior median plus a credible band.
- Wide where the data is uninformative.
- Honest about how sure the ratio really is.

The Hedge Ratio With a Credible Band

The posterior median hedge ratio, wrapped in a 95% credible band from the FFBS draws. The band widens when the data says less — exactly the honesty a single filtered line could never provide.



Trading Stage III, Now Honest

The band changes the signal

With the old line

- You used one hedge ratio, taken as exact.
- The spread it implied looked precise.
- You couldn't tell a firm signal from a shaky one.

With the band

- A tight band means a trustworthy hedge ratio.
- A wide band warns the relationship is uncertain.
- You can stand down when the pair is ill-defined.

TVP-SV: Adding Stochastic Volatility

When the noise level is itself a hidden, moving process

So far the noise level was constant. But financial volatility clusters and drifts. TVP-SV lets the variance itself be a hidden, evolving state — a time-varying coefficient on top, a breathing volatility underneath, both inferred at once.

The limitation so far

- We assumed a fixed observation-noise level.
- But market volatility is anything but fixed.
- It clusters — calm spells and turbulent spells.

The TVP-SV answer

- Add a second hidden state: the log-volatility.
- Let it drift like the coefficient does.
- Infer both moving processes jointly.

Why Constant Variance Isn't Enough

Volatility is a moving target

What constant variance misses

- It averages calm and crisis into one level.
- It understates risk in turbulent periods.
- And overstates it in quiet ones.

What a moving variance captures

- Bands that widen in turbulent stretches.
- Tighter inference when markets are calm.
- Risk estimates that track the real environment.

The TVP-SV Model

Two hidden processes, one system

$$\begin{aligned} y_t &= \beta_t x_t + e^{h_t/2} \varepsilon_t \\ \beta_t &= \beta_{t-1} + \eta_t, \quad h_t = h_{t-1} + \nu_t \end{aligned}$$

The two states

- The coefficient drifts as a random walk.
- The log-volatility drifts as its own random walk.
- The observation noise scales with that volatility.

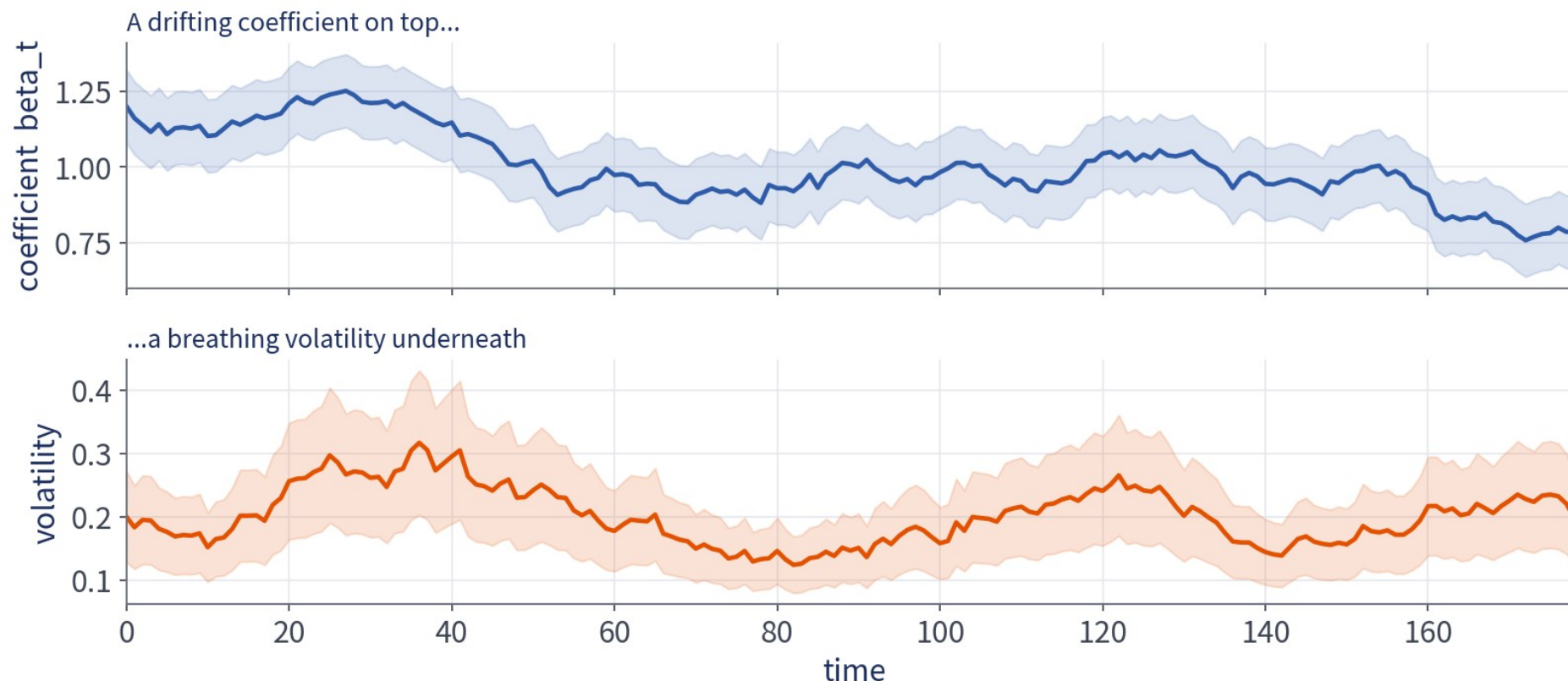
Inferring it

- The same Gibbs strategy, with an extra block.
- A dedicated sampler handles the volatility state.
- Out comes a joint posterior over both processes.

TVP-SV in One Picture

A drifting coefficient on top and a breathing volatility underneath, each with its own credible band. Both are hidden states, inferred together — the model adapts its own sense of risk over time.

TVP-SV: both the coefficient AND the noise level are hidden, moving processes



Bayesian Time-Varying Hedge Ratio

Direct Claude Code to fit a dynamic linear model in PyMC

GOAL Fit a drifting hedge ratio as a Bayesian state-space model and produce a credible band.

[INPUT] Prompt · Claude Code

Write and run `bayes_hedge.py` that:

1. simulates a pair whose hedge ratio drifts as a random walk (a local-level DLM);
2. specifies a Bayesian state-space model in PyMC with priors on the state and observation noise;
3. samples the joint posterior over the whole hedge-ratio path and the parameters;
4. plots the posterior median hedge ratio with a 95% credible band against the true path.

Show the band widens where the data is less informative and covers the truth.

How to work with it

- The hedge ratio is the hidden state.
- Priors go on the two noise levels.
- The output is a fan, not a line.
- Wide band = shaky hedge ratio.

Deliverable: `bayes_hedge.py` + a posterior-median hedge ratio with a credible band vs the truth.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What are the two blocks in the Gibbs sweep?

**States given params,
params given states**

Q2 What does one FFBS sweep produce?

**One full sampled
state path**

Q3 What extra hidden process does TVP-SV add?

**A drifting log-
volatility**

All three come from the two-block sweep, FFBS, and the stochastic-volatility state.

Three Pitfalls to Avoid

Where the sampling goes wrong

Common mistakes

- "I can draw each state independently from the filter."
- "FFBS gives a smoothed line."
- "Constant variance is fine for financial data."

The reality

- Independent per-time draws are inconsistent — use FFBS.
- FFBS draws a full sampled PATH, not a mean.
- Volatility drifts; TVP-SV models it as a hidden state.

Three Things to Carry Forward

Part 2 in one screen

1

Alternate states and parameters

A two-block Gibbs sweep draws the whole state path given the parameters, then the parameters given the states. Each conditional is easy; alternating explores the joint posterior.

2

FFBS draws a whole path at once

Forward-filter to gather information, then sample backward from the last state to the first. Each sweep yields one coherent path; many sweeps make the fan and its credible band.

3

The hedge ratio becomes a fan, and volatility can breathe

The Stage III hedge ratio gains an honest credible band. TVP-SV adds a drifting volatility, so the model tracks changing risk as a hidden state of its own.

Next — Part 3: the Bayesian HMM, soft regime probabilities, structural breaks, and pairs Stage VI.

★ WEEK 15 · PART 2 · PRACTICE

Hedging and Risk, Worked by Hand

A Dynamic Hedge Ratio and a Breathing VaR

Compute h from five days; update volatility and VaR by hand

Part 2 at work, with the arithmetic shown. We compute a minimum-variance hedge ratio from five days, measure the variance it removes, update volatility with one shock, and turn it into a Value-at-Risk that breathes.

2P

PART 2 · APPLICATIONS

Two Desks, Computed by Hand

Case A: compute the hedge ratio h from five days, see how much risk it cuts, then update it. Case B: update volatility with one big return, then read off a VaR that a fixed limit would miss. Then a Claude Code lab.

What Part 2 Gave Us

Now we put them to work - with numbers

1

FFBS draws whole state paths

A two-block Gibbs sweep alternates states and parameters; FFBS samples a full path per sweep, giving a median with a credible band.

2

The hedge ratio becomes a fan

A drifting coefficient, inferred as a hidden state, turns one hedge number into a band. We first compute that number by hand, then update it.

3

TVP-SV lets volatility breathe

A stochastic-volatility state makes the noise level move. We approximate it by hand with one EWMA update, then read off the VaR.

Everything here is a calculator exercise: five days of hedging data, one volatility update, one VaR.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key number by hand, step by step.
- Then update it as new data arrives.

What you should watch

- How much risk the hedge actually removes.
- How volatility jumps after one shock.
- How a fixed limit misses that jump.

The Dynamic Futures Hedge

A hedge ratio that must track a drifting optimum

A firm hedges a spot position with futures. The minimum-variance hedge ratio is $\text{Cov}(\text{spot}, \text{futures})$ over $\text{Var}(\text{futures})$. We compute it from five days, measure how much variance it removes, then update it when the next window disagrees.

The setup

- A spot position to protect with futures.
- The best hedge is the minimum-variance ratio.
- That ratio shifts as correlations move.

What we'll do by hand

- Compute h from five days.
- Measure the variance it removes.
- Update h for a new window.

The Minimum-Variance Hedge Ratio

And why it drifts

$$h_t^* = \frac{\text{Cov}(\Delta S, \Delta F)}{\text{Var}(\Delta F)} \text{ estimated as a drifting regression coefficient } \beta_t$$

What it is

- The covariance of spot and futures over the futures variance.
- The slope that minimizes hedged variance.
- Equivalently, a regression coefficient.

Why model it dynamically

- That covariance is not constant.
- So the optimal ratio drifts.
- A random-walk coefficient captures it.

The Data: Five Days

Compute the hedge ratio h by hand

Day	Futures dF (%)	Spot dS (%)
1	+1.0	+1.2
2	-0.8	-0.5
3	+1.5	+0.9
4	-1.2	-1.0
5	+0.6	+0.9

Goal: $h = \text{Cov}(\text{spot}, \text{futures}) / \text{Var}(\text{futures})$. We compute it in three lines on the next slide.

$h = \text{Cov} / \text{Var}$

Three lines with a calculator

$$\begin{aligned}\overline{\Delta F} &= 0.22, & \overline{\Delta S} &= 0.30 \\ \text{Cov}(\Delta S, \Delta F) &= \frac{1}{5} \sum (\Delta F - \overline{\Delta F})(\Delta S - \overline{\Delta S}) = \frac{4.36}{5} = 0.872 \\ \text{Var}(\Delta F) &= \frac{1}{5} \sum (\Delta F - \overline{\Delta F})^2 = \frac{5.448}{5} = 1.090 \\ h^* &= \frac{\text{Cov}}{\text{Var}} = \frac{0.872}{1.090} = \mathbf{0.80}\end{aligned}$$

The minimum-variance hedge is $h = 0.80$: short 0.8 futures for each unit of spot. Now, how much risk does that remove?

How Much Risk It Removes

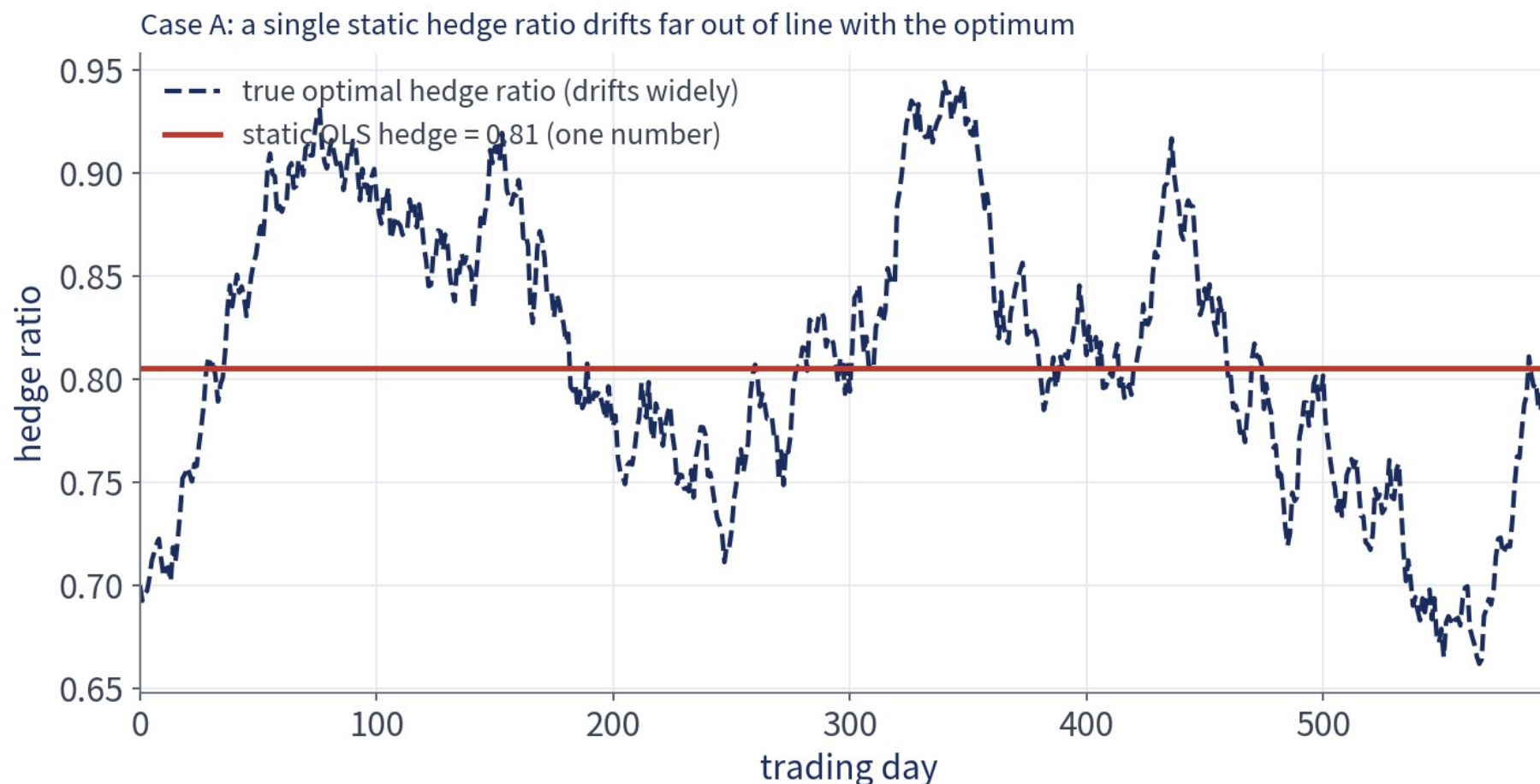
The whole point of a hedge

$$\begin{aligned}\text{unhedged: } \text{Var}(\Delta S) &= 0.77 \\ \text{hedged: } \text{Var}(\Delta S - 0.80 \Delta F) &= 0.074 \\ \text{risk cut} &= 1 - \frac{0.074}{0.77} \approx \mathbf{90\%} \text{ less variance}\end{aligned}$$

Hedging with $h = 0.80$ cuts the position's variance by about 90%. A wrong h would leave far more residual risk.

Static Hedge vs the Drifting Optimum

Compute that hedge ratio once over the whole sample and you get one flat number. But the true optimum drifts widely: the static hedge is right on average and wrong most of the time - over- and under-hedging



Update the Hedge Ratio

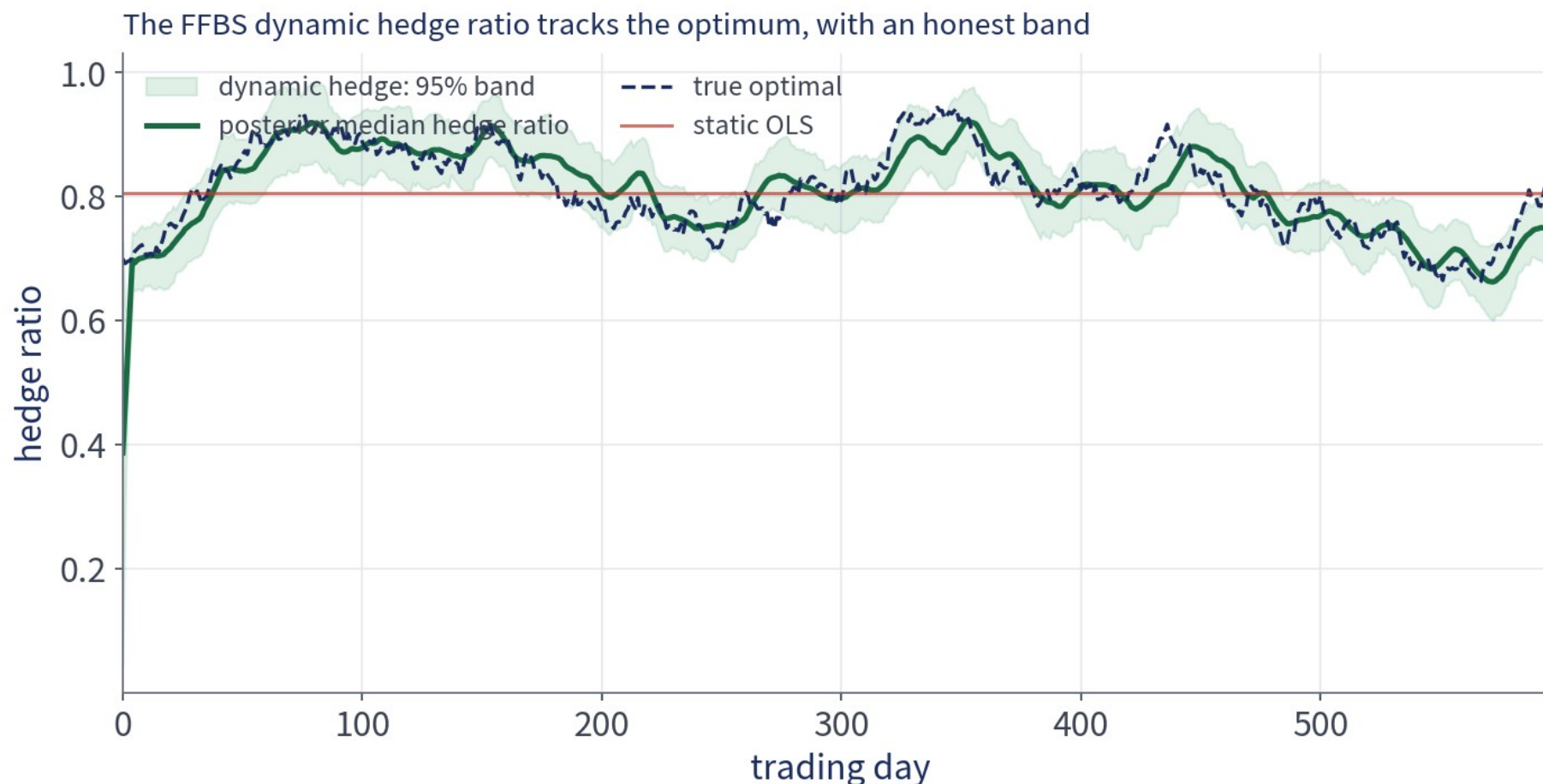
Prior + new window = posterior, by hand

$$\begin{aligned}\text{prior } h &\sim N(0.80, 0.15^2), & \text{new window } y &= 1.00 \pm 0.10 \\ \text{precision} &= \frac{1}{0.15^2} + \frac{1}{0.10^2} = 44.4 + 100 = 144.4 \\ \text{posterior mean} &= \frac{44.4(0.80) + 100(1.00)}{144.4} = \frac{135.6}{144.4} = \mathbf{0.94} \pm 0.083\end{aligned}$$

The updated hedge is a precision-weighted average: 0.94, pulled toward the sharper new window, with its own band of 0.083.

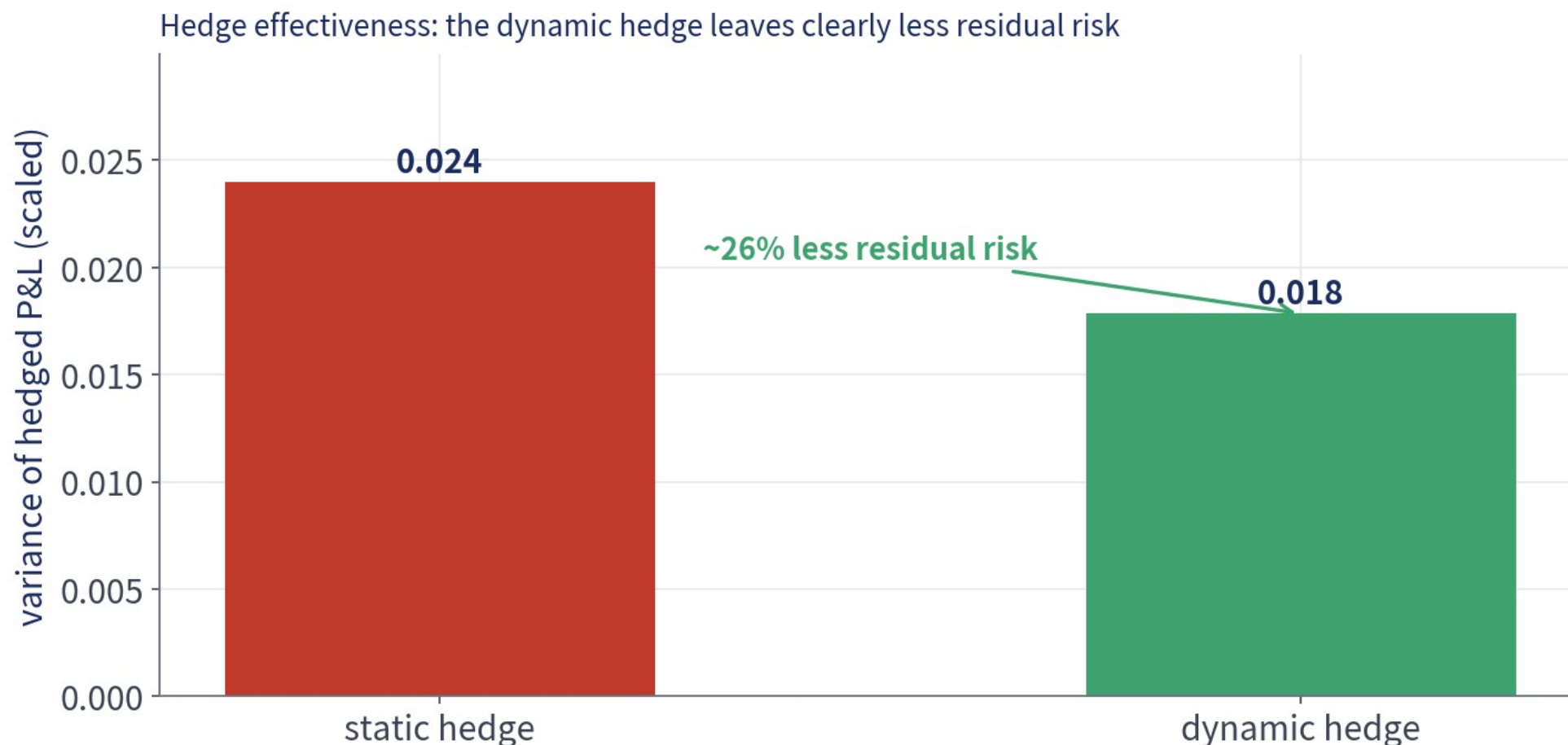
The FFBS Dynamic Hedge Ratio

Repeating that update every day gives a hedge ratio that tracks the drifting optimum and carries a credible band. The band widens where the relationship is noisy - an honest signal to hedge more cautiously



The Payoff: Less Residual Risk

Over the full sample, the same story as the hand calculation: because it stays aligned with the drifting optimum, the dynamic hedge leaves clearly less residual variance than the single static number.



What the Stale Hedge Ratio Costs

Basis risk you didn't sign up for

With the static ratio

- The hedge is mis-sized whenever the optimum moved.
- Residual variance leaks back into the position.
- You believe you are hedged more than you are.

With the dynamic ratio

- The hedge follows the optimum as it drifts.
- Residual risk stays low through the changes.
- The band flags when the hedge is less reliable.

Value-at-Risk That Breathes

A risk limit built on one volatility number

A risk desk sets a Value-at-Risk from a volatility estimate. We update volatility by hand with one big return using EWMA, watch it jump, then compute the VaR - and find a fixed limit that a real loss would breach while the breathing one holds.

The setup

- A daily VaR limit for a trading book.
- It scales directly with estimated volatility.
- One volatility number sets one fixed limit.

What we'll do by hand

- Update volatility after a shock (EWMA).
- Turn each volatility into a VaR.
- Find the loss that breaches the fixed limit.

A Value-at-Risk That Moves

Volatility as a hidden, breathing state

$$\text{VaR}_t = z_\alpha \sigma_t, \quad \sigma_t = e^{h_t/2}, \quad h_t = h_{t-1} + \nu_t \quad (\text{a hidden, moving level})$$

What it says

- VaR is a multiple of the current volatility.
- That volatility is the exponential of a log-vol state.
- The log-vol drifts as its own random walk.

Our hand version

- We approximate the moving vol with EWMA.
- It rises after a shock and decays in calm.
- Then VaR is just 1.645 times that vol.

Volatility After One Shock

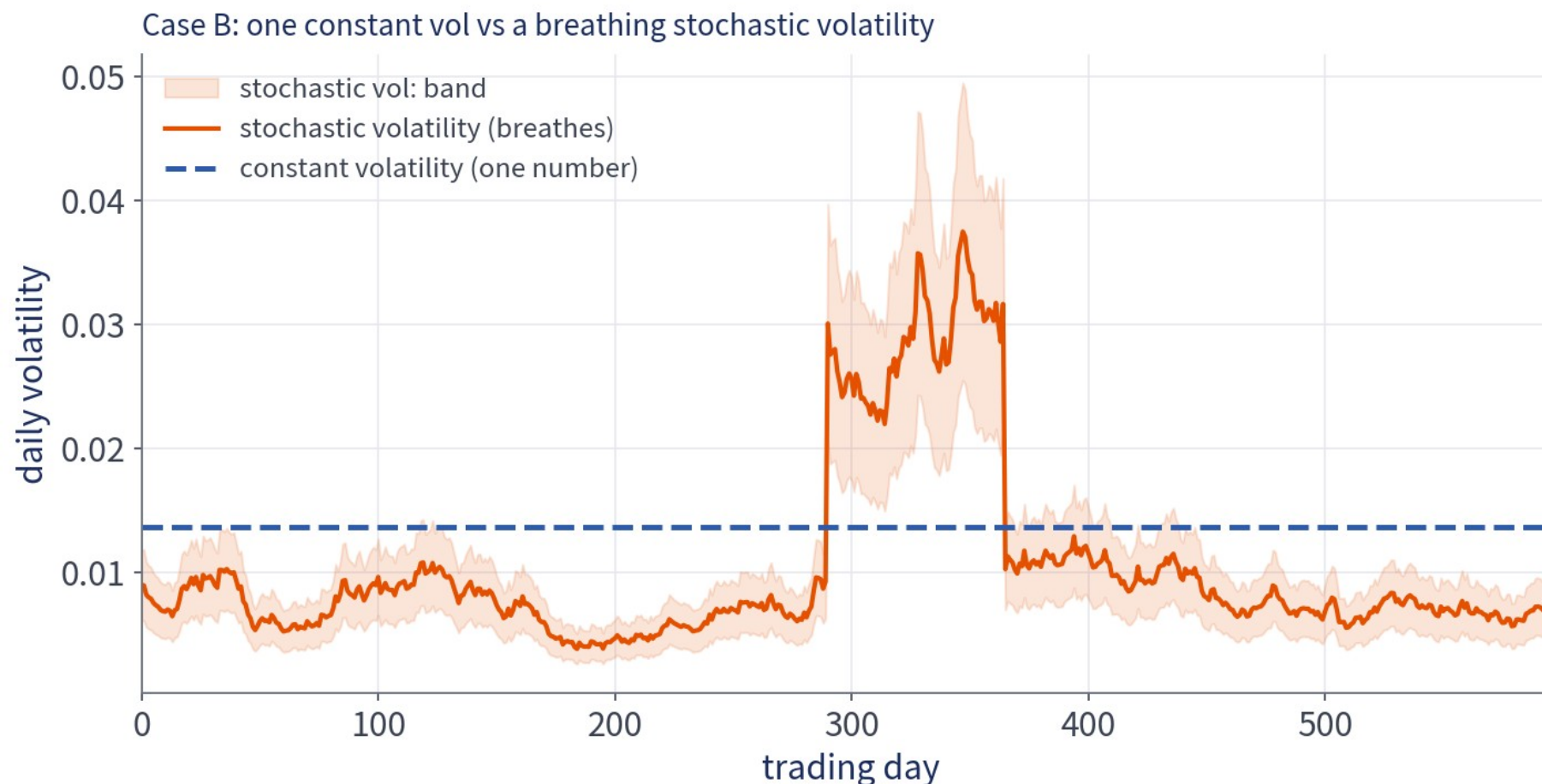
The EWMA update, by hand

$$\begin{aligned}\sigma_t^2 &= \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94 \\ &= 0.94(0.010)^2 + 0.06(-0.040)^2 \\ &= 0.000094 + 0.000096 = 0.000190 \\ \sigma_t &= \sqrt{0.000190} = \mathbf{1.38\%} \text{ (up from 1.00\%)}\end{aligned}$$

One 4% return drives volatility from 1.00% to 1.38% in a single step. This is how the estimate 'breathes' after stress.

Constant vs Stochastic Volatility

Do that update every day and volatility breathes: quiet for long stretches, then spiking in the stress window - exactly the jump we just computed by hand. A single constant volatility is one flat line through it all



VaR, and a Breach

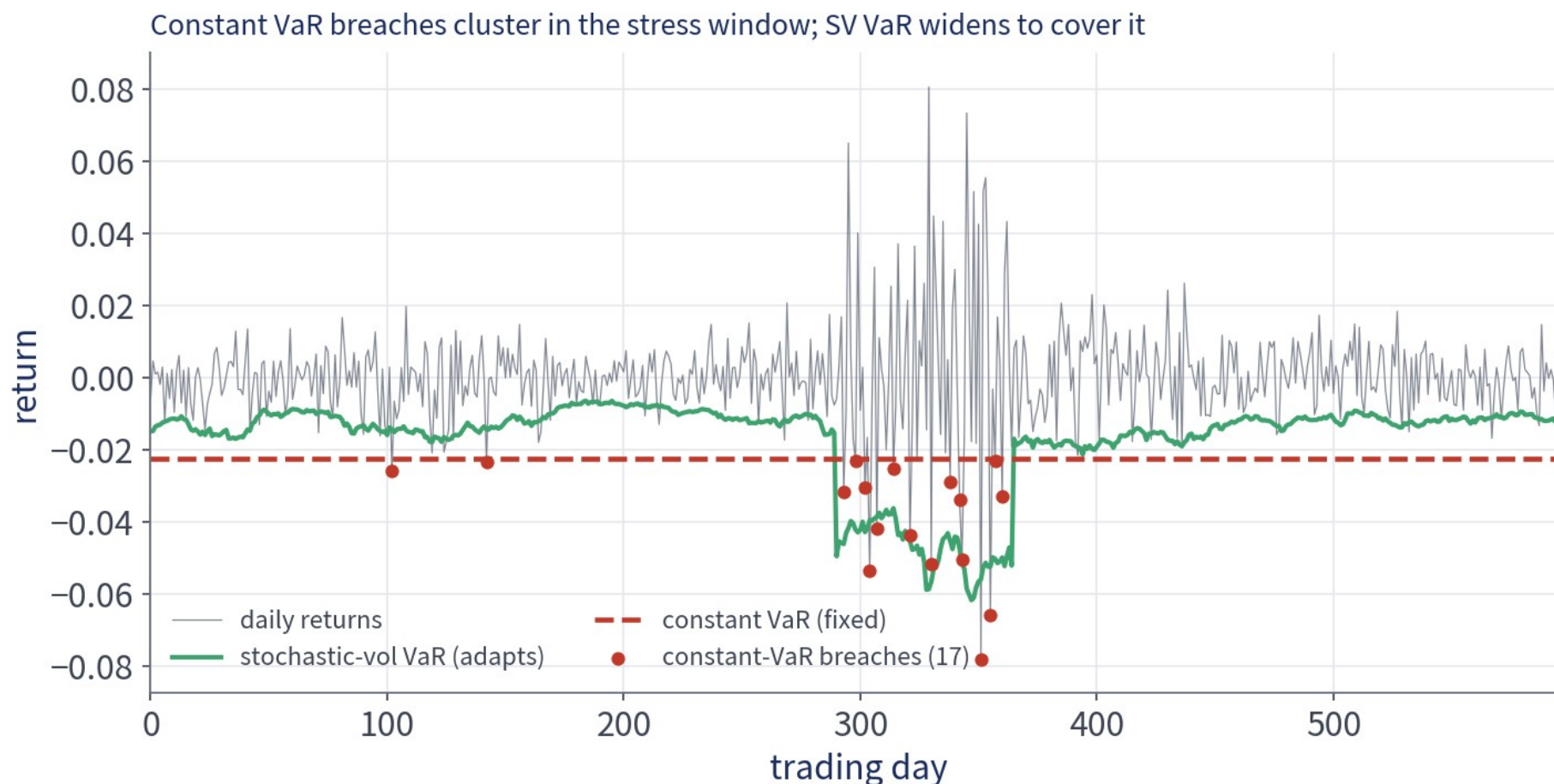
Turn volatility into a limit

$$\begin{aligned} \text{VaR}_{95\%} &= 1.645 \sigma \\ \text{calm: } 1.645(1.00\%) &= 1.65\%, & \text{after shock: } 1.645(1.38\%) &= 2.27\% \\ \text{constant (avg } 1.2\%) : 1.645(1.2\%) &= 1.97\% \\ \text{a } -2.1\% \text{ day: } \underbrace{2.1\% > 1.97\%}_{\text{breaches constant}}, & \underbrace{2.1\% < 2.27\%}_{\text{covered by SV}} \end{aligned}$$

A 2.1% loss breaches the fixed 1.97% limit but sits inside the post-shock 2.27% limit. The breathing VaR saw the storm coming; the fixed one didn't.

Constant VaR Breaks in Clusters

Over a full path, the fixed VaR is breached again and again in the stress window - precisely when it matters. The stochastic-vol VaR widens to cover the storm, so its breaches stay rare and spread out.



The Quant-Quake Echo

When a stale risk number turns procyclical

How it snowballs

- A fixed limit is fine until volatility jumps.
- Then clustered breaches force sudden deleveraging.
- Forced selling raises volatility further.

The honest alternative

- A breathing limit tightens gradually as vol rises.
- Adjustments are smaller and earlier.
- Less of the abrupt, cascading unwind.

The Common Pattern

Two desks, one moving quantity

Both cases, now computed by hand, replace a frozen number with a state that moves. The hedge ratio drifts; the volatility breathes. In each, updating the estimate - a weighted average, an EWMA step - resizes the decision as the market changes.

The trap

- A quantity that should move, held fixed.
- A hedge or a limit pinned to that stale value.
- A failure that arrives at the worst time.

The fix

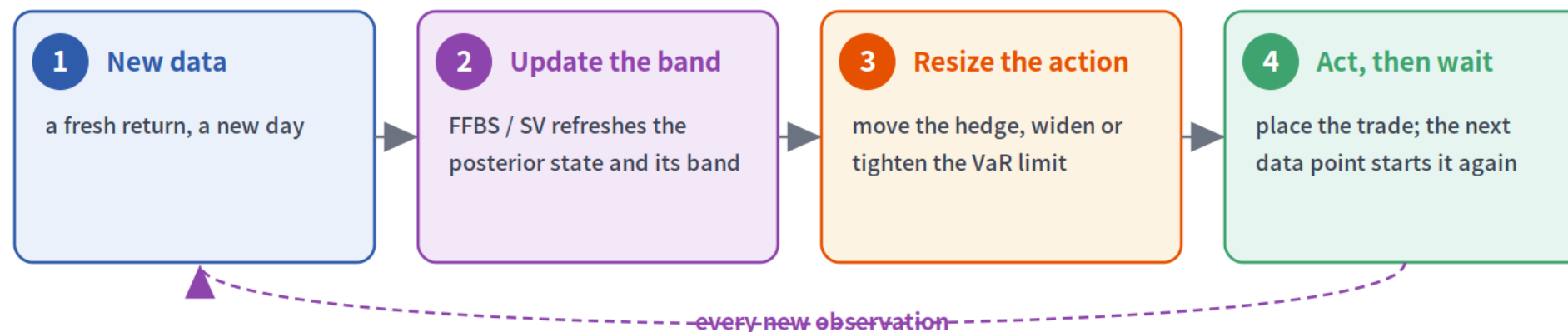
- Update the quantity as data arrives.
- Carry a band around it.
- Resize the hedge or limit as it moves.

A Decision That Moves With the Market

New data updates the estimate and its band - a weighted average for the hedge, an EWMA step for the vol; the band resizes the hedge or the risk limit; the next observation starts the loop again. The band is the input, not a report

A Decision That Moves With the Market

New data updates the posterior band; the band resizes the hedge or the risk limit. Then repeat.



Static once, or adaptive always

Case A resizes the hedge as the hedge ratio drifts; Case B resizes the VaR limit as volatility breathes. Both close the same loop: the posterior band is not a report - it is the input to the next decision.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full time series.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - hedge:

- * take the 5-day `dF`, `dS` table; compute `Cov`, `Var`, and `h`, printing each number;
- * compute `Var(dS)` and `Var(dS - h*dF)`; report the percent risk reduction;
- * do the normal-normal update of `h` by hand-formula.

CASE B - VaR:

- * apply one EWMA update: $0.94 \cdot \text{sig}^2 + 0.06 \cdot r^2$;
- * print the new volatility;
- * compute calm, shock, and constant VaR at 1.645;
- * show which VaR a 2.1% loss breaches.

Match every printed number to the slides.

Check against the slides

- $h = 0.80$; risk cut $\sim 90\%$.
- Updated $h = 0.94 \pm 0.083$.
- Vol $1.00\% \rightarrow 1.38\%$ after shock.
- 2.1% loss breaches 1.97%, not 2.27%.

Deliverable: `part2_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get the hedge ratio from the five-day table?

$\text{Cov}(\text{spot}, \text{futures}) / \text{Var}(\text{futures})$

Q2 In one EWMA step, what makes volatility jump?

The large squared return term

Q3 Why does a 2.1% loss breach the fixed VaR but not the SV VaR?

The SV VaR widened after the shock

All three come from the hedge calc, the EWMA update, and the two VaR numbers.

Two Calculations to Remember

Part 2 applications in one screen

1

A hedge ratio is Cov/Var - and it cuts real risk

We computed $h = 0.80$ from five days, which removed about 90% of the position's variance. A static number drifts out of line; updating it (to 0.94) keeps the hedge aligned as the market changes.

2

Volatility breathes, and VaR must follow

One EWMA step took volatility from 1.00% to 1.38%. A 2.1% loss breaches the fixed 1.97% limit but not the post-shock 2.27% one - the breathing VaR saw the storm the fixed one missed.

3

The update is the input, not the output

In both desks, each update - a weighted average, an EWMA step - feeds the next decision, resizing the hedge and the limit as the market moves. That is what makes the response adaptive.

Next - Part 3 practice: soft regimes and structural breaks, also worked by hand.

★ WEEK 15 · PART 3 of 3

Bayesian Switching: Regimes, Breaks, and Stage VI

Soft Regime Calls and Trading With Uncertainty

Bayesian HMM · structural breaks · label switching · the capstone

We re-estimate the regime model of Week 10 with priors, get soft credible regime probabilities instead of hard labels, locate structural breaks at uncertain dates, and complete the pairs-trading thread at Stage VI.

3

PART 3 · BAYESIAN SWITCHING

Regimes, Breaks, and the Capstone

The Bayesian HMM and soft regime probabilities, structural breaks at uncertain dates, the label-switching gotcha, sizing positions by conviction, and pairs trading Stage VI.

From Drifting States to Switching States

Parts 1-2 drifted; Part 3 switches

1

We handled continuous drift

Parts 1-2: hedge ratios and volatilities that drift smoothly, with FFBS drawing whole paths and honest bands.

2

But markets also switch abruptly

Calm flips to crisis; a relationship breaks at a date. These are discrete regimes and jumps, not smooth drift.

3

Bayes handles switching too

The Bayesian HMM gives soft regime probabilities; break models locate jumps with uncertainty. Both feed position sizing.

Part 3 in one line: soft regime probabilities and uncertain break dates, flowing through to how much you trade.

The Bayesian HMM

Re-estimating Week 10's regime model with priors

Week 10 fit a hidden Markov model by Baum-Welch — a point estimate of the transition matrix and the regime parameters. The Bayesian HMM puts priors on all of it and infers a full posterior, so every regime quantity comes with honest uncertainty.

What Week 10 gave

- A single best transition matrix.
- Point estimates of each regime's parameters.
- Hard-ish regime assignments via the filter.

What Bayes adds

- Priors on the transitions and emissions.
- A posterior over transitions, parameters, and states.
- Credible bands on everything, regimes included.

The Bayesian HMM Posterior

Transitions, emissions, and regimes together

$$p(\underbrace{A}_{\text{transitions}}, \underbrace{\phi}_{\text{emissions}}, \underbrace{s_{1:T}}_{\text{regimes}} \mid y) \propto p(y \mid s, \phi) p(s \mid A) \underbrace{p(A)}_{\text{Dirichlet}} p(\phi)$$

The pieces

- The transition matrix, with a Dirichlet prior on each row.
- The emission parameters for each regime.
- The hidden regime sequence itself.

How it's sampled

- The same Gibbs strategy as the drift models.
- Draw the regime path, then the parameters, and repeat.
- FFBS has a discrete cousin for the regime path.

From Baum-Welch to a Posterior

A point fit becomes a distribution

The old fit

- Baum-Welch maximizes the likelihood.
- It returns one transition matrix and one set of parameters.
- Uncertainty about them is invisible.

The Bayesian version

- Priors regularize small-sample regime estimates.
- The posterior spreads over plausible transition matrices.
- That spread flows into the regime probabilities.

Bayesian Regime Detection

Credible regime probabilities, not hard labels

The point of the Bayesian HMM is a softer, more honest answer to 'what regime are we in?'. Instead of a hard label that flips back and forth, you get a probability of crisis at each date — with a credible band around it.

The hard-label problem

- A threshold turns probabilities into a 0/1 label.
- Near the threshold, it flip-flops noisily.
- It looks decisive but is often just noise.

The soft alternative

- Report the probability of each regime directly.
- Add a credible band from the posterior.
- Decisiveness now reflects real evidence.

Soft Probability vs Hard Label

Two ways to answer the regime question

A hard label says

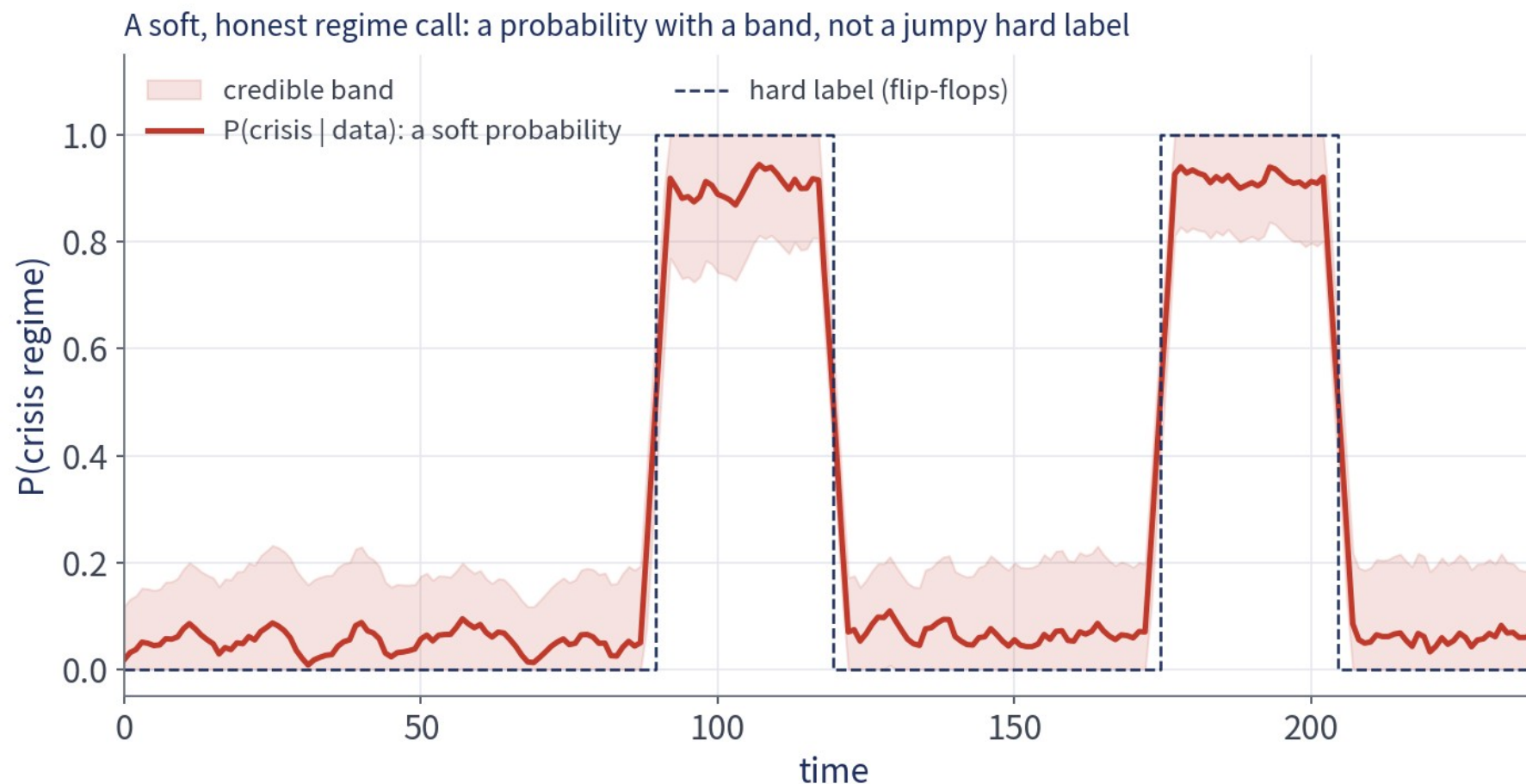
- 'We are in a crisis' — or not.
- One bit, no sense of confidence.
- It jumps whenever the estimate crosses a line.

A soft probability says

- 'There is a 70% chance we are in a crisis.'
- It carries its own confidence.
- It moves smoothly as evidence accumulates.

A Soft, Honest Regime Call

The probability of crisis, as a smooth posterior curve with a credible band, versus a hard 0/1 label that flip-flops. The soft call tells you not just which regime, but how sure the model is.



Why Soft Probabilities Beat Hard Labels

Especially at the edges

For understanding

- You see the ramp into and out of a regime.
- Ambiguous periods look ambiguous, not certain.
- You are never fooled by a confident-looking flip.

For decisions

- You can scale exposure by the probability.
- No arbitrary threshold to trip over.
- Uncertainty flows straight into position size.

Structural Break Points

When the parameters jump at an unknown date

Sometimes the world doesn't cycle between regimes — it changes once and stays changed. A structural break is a one-time jump in the parameters at an unknown date. The Bayesian approach puts a posterior on WHEN the break happened.

What a break is

- The model's parameters shift at a single date.
- Before and after look like different regimes.
- But it happens once, not repeatedly.

The Bayesian question

- Treat the break date as an unknown parameter.
- Infer a posterior over when it occurred.
- Get both whether AND when, with uncertainty.

The Break Date as an Unknown

Put a posterior on 'when'

The idea

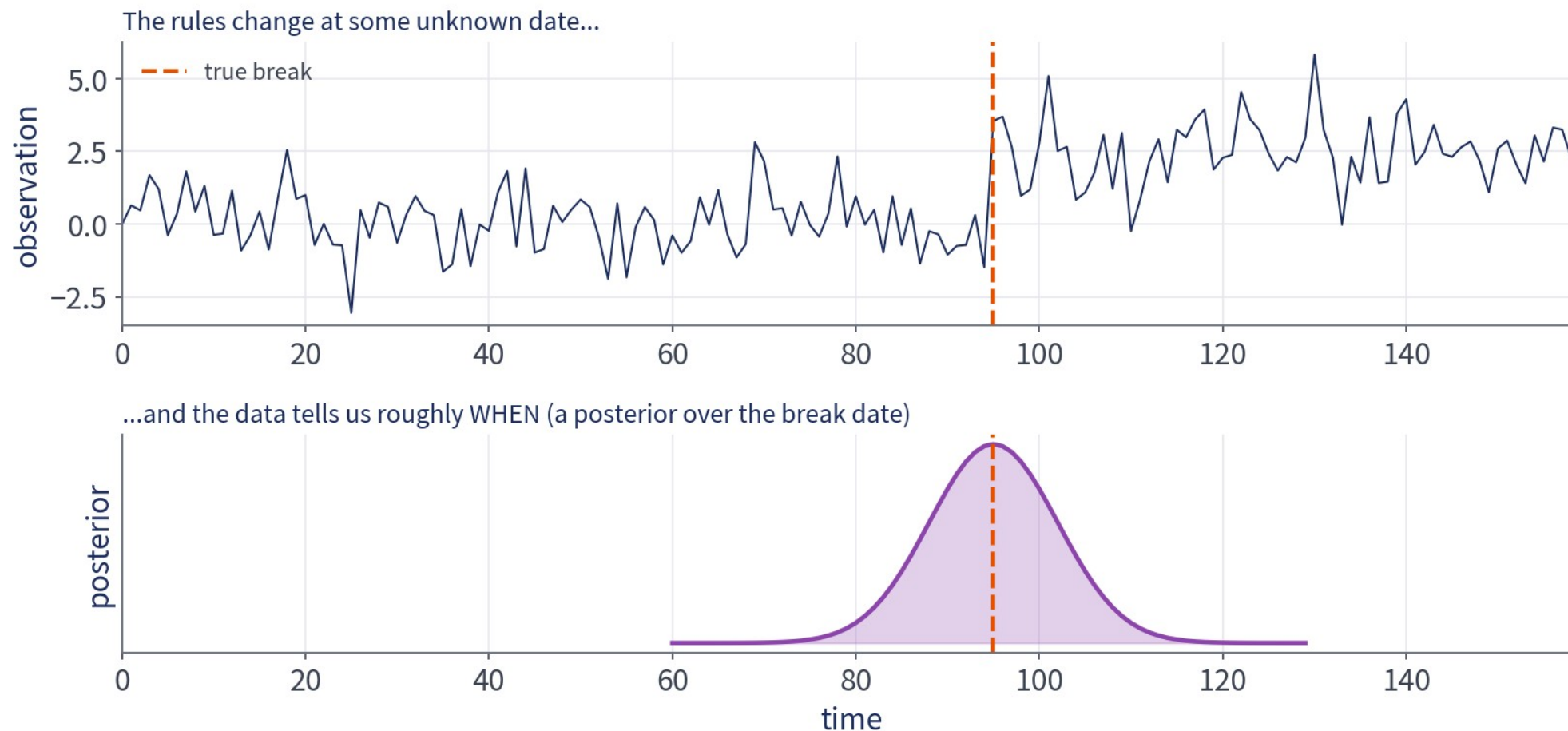
- The change-point is just another parameter.
- The data is more likely under the right split.
- So the posterior concentrates near the true date.

What you learn

- A most-probable break date — and a range around it.
- Whether the break is sharp or the date is fuzzy.
- Honest uncertainty about the timing itself.

A Structural Break With an Uncertain Date

The series shifts at some unknown time (top). The data tells us roughly when: a posterior over the break date (bottom) peaks near the true change but spreads to reflect our uncertainty about exactly when.



Regimes vs Breaks

Recurring states or a one-time jump

Regimes (HMM)

- The world cycles among a few recurring states.
- Calm and crisis come and go, and come back.
- Modeled by a hidden Markov chain.

Structural breaks

- A single, permanent shift in the rules.
- No return to the old regime.
- Modeled by a change-point at an unknown date.

A Bayesian HMM Gotcha: Label Switching

The regimes have no inherent names

A subtle trap: the model doesn't know which regime is 'regime 1'. Swapping the labels leaves the likelihood unchanged, so during sampling the chain can relabel the regimes. Average the raw draws blindly, and the regimes blur into each other.

Why it happens

- The likelihood is identical under any relabeling.
- So 'regime 1' and 'regime 2' can trade places mid-chain.
- The names are arbitrary; the model can't tell them apart.

The symptom

- Averaging a regime's parameter mixes both regimes.
- The two regimes collapse toward the same values.
- Your posterior looks meaningless.

Handling Label Switching

Pin the labels down

Simple remedies

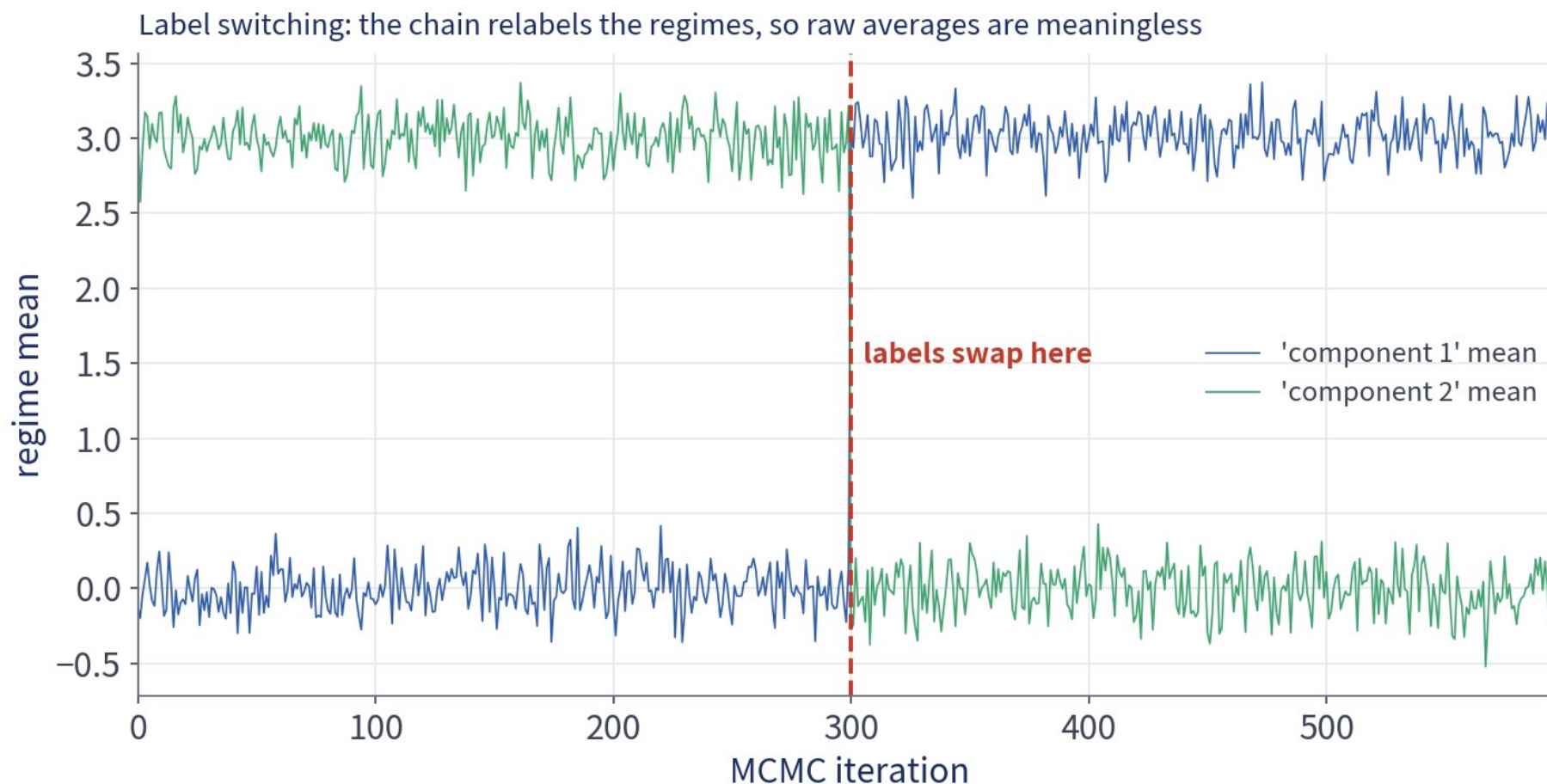
- Impose an ordering, e.g. regime means increasing.
- That breaks the symmetry and fixes the names.
- Often enough for two or three regimes.

More careful remedies

- Relabel the draws after sampling to align them.
- Then summarize each regime consistently.
- Always check the traces for swaps before trusting summaries.

Label Switching in Action

The trace of 'component 1's' mean jumps between two values partway through: the chain has relabeled the regimes. A naive average would land between them and describe neither regime correctly.



From Posteriors to Position Sizes

Letting uncertainty flow all the way through

All this honesty matters only if it changes what you do. The final step lets the posterior flow into the trade: a strong edge with a tight band and a confident regime gets full size; a shaky edge with a wide band or an ambiguous regime gets scaled down.

What feeds the size

- The strength of the signal (the edge).
- The conviction: how probable a favorable regime is.
- The tightness of the credible band.

The result

- Big when sure, small when doubtful.
- No hard on/off switch to whipsaw you.
- Risk scaled to real, stated uncertainty.

Sizing by Conviction

One expression for 'size to your certainty'

$$\text{position}_t \propto \underbrace{\text{edge}_t}_{\text{signal}} \times \underbrace{P(\text{favorable} \mid y)}_{\text{conviction}} \times \underbrace{\frac{1}{\text{band width}}}_{\text{less doubt, more size}}$$

Reading it

- Start from the raw edge — the signal.
- Scale by the probability the regime is favorable.
- Shrink when the credible band is wide.

The behavior

- Full size only when signal, regime, and band all agree.
- Automatic caution when any of them is shaky.
- Uncertainty is not a footnote — it sets the bet.

Pairs Trading Stage VI: Trading With Uncertainty

The thread's completion in the core course

This is where the pairs-trading thread arrives. Stage VI trades the pair using everything: a hedge ratio with a credible band, a soft regime probability, and conviction-based sizing. You trade by how sure you are — not by a hard label.

What Stage VI uses

- The Bayesian hedge ratio and its band (Part 2).
- The soft regime probability (this part).
- Conviction-based position sizing.

How it trades

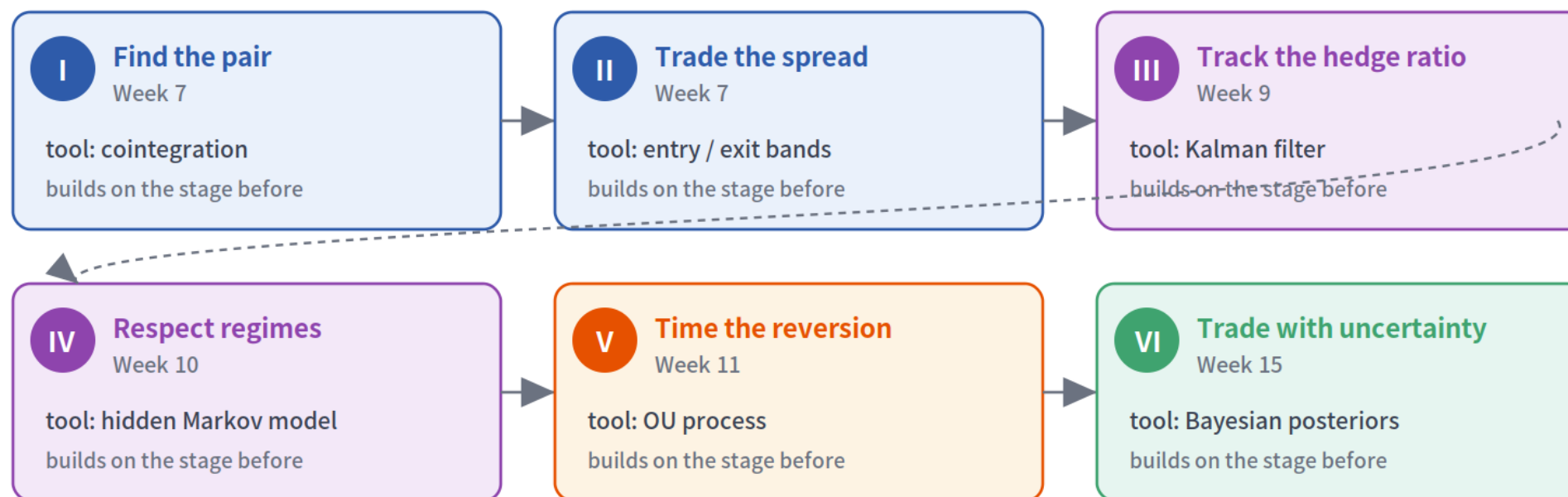
- Enter firmly when everything aligns and is tight.
- Trade small, or stand aside, when doubtful.
- Uncertainty is built into every decision.

The Pairs-Trading Thread, Assembled

Six stages, one evolving strategy: find the pair, trade the spread, track the drifting hedge ratio, respect regimes, time the mean reversion, and finally trade with honest uncertainty. Stage VI completes the core course.

The Pairs-Trading Thread, Assembled

Six stages, one evolving strategy - Stage VI completes the core course.



Stage VI adds the missing ingredient - honest uncertainty - and closes the core course.

Stage VI in Practice

Trade by conviction, not by a label

The old way

- A hard signal said in or out.
- One hedge ratio, taken as exact.
- The same size regardless of confidence.

The Stage VI way

- A probability of a good setup, not a yes/no.
- A hedge ratio with a band you can see.
- Size that rises and falls with conviction.

Bayesian Regimes for the Pair (Stage VI)

Direct Claude Code to trade by conviction

GOAL Fit a Bayesian regime model to the pair and size the trade by conviction, not a hard label.

[INPUT] Prompt · Claude Code

Write and run stage6_pair.py that:

1. fits a two-regime Bayesian HMM to a pair's spread (calm vs stressed), with priors;
2. imposes an ordering to avoid label switching;
3. outputs a soft $P(\text{favorable regime})$ with a band, not a hard 0/1 label;
4. sizes the position by edge times conviction, scaled down when the band is wide.

Compare cumulative results to a hard-label rule and show the conviction-sized version is steadier.

How to work with it

- Order the regimes to fix labels.
- Report probabilities, not labels.
- Size by conviction and band width.
- Smaller when the model is unsure.

Deliverable: stage6_pair.py + soft regime probabilities and conviction-sized results vs a hard rule.

Three Questions Before You Finish

If you can answer these, Week 15 has landed

Q1	What does a Bayesian HMM give instead of a hard regime label?	A soft probability with a band
Q2	What is label switching?	The chain relabels the regimes
Q3	What three things set the Stage VI position size?	Edge, conviction, band width

All three come from soft regime calls, the label-switching gotcha, and conviction-based sizing.

Three Pitfalls to Avoid

Where Bayesian switching goes wrong

Common mistakes

- "A hard regime label is the honest answer."
- "I can average the raw HMM draws directly."
- "A structural break happened on a known date."

The reality

- A soft probability with a band is more honest.
- Fix label switching first, or the regimes blur together.
- The break date is itself uncertain — infer it.

Week 15 in Three Ideas

Pulling the whole week together

1

Point estimates hide uncertainty; posteriors reveal it

Parts 1-2: a filtered line omits parameter uncertainty. A Bayesian state-space model, sampled with FFBS, turns it into an honest fan — a hedge ratio and even a volatility with credible bands.

2

Switching, done honestly

Part 3: the Bayesian HMM gives soft regime probabilities, not flip-flopping labels; break models locate jumps at uncertain dates. Mind label switching before summarizing.

3

Uncertainty flows to the trade

Conviction-based sizing scales exposure by edge, regime probability, and band width. Pairs Stage VI trades by how sure you are — completing the thread and the core course.

Next — Week 16: Synthesis and the Final Project, where the whole course comes together.

★ WEEK 15 · PART 3 · PRACTICE

Regimes and Breaks, Worked by Hand

Soft Regime Probabilities and Conviction Sizing

One Bayes step for a regime; SSE splits for a break

Part 3 at the trading desk, with the arithmetic shown. We turn a single return into a regime probability with one Bayes step, size the trade by conviction, then locate a structural break by comparing sums of squares.

3P

PART 3 · APPLICATIONS

Two Trading Decisions, by Hand

Case A: turn one -3% day into $P(\text{stress})$ with a Bayes step, then size by conviction instead of a hard label. Case B: find when a relationship broke by comparing split fits. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

Soft regime probabilities beat hard labels

A Bayesian HMM reports the probability of each regime, not a flip-flopping 0/1 label. We compute one such probability by hand from a single day.

2

Breaks happen at uncertain dates

When the rules change once, the data tells us roughly when. We locate that date by comparing how well different splits fit.

3

Uncertainty flows to position size

Conviction sizing scales the trade by the regime probability. We compute soft sizes and contrast them with a hard on/off rule.

Everything here is a calculator exercise: one Bayes update, a few sums of squares, one sizing rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key number by hand, step by step.
- Then turn it into a trading decision.

What you should watch

- How one day updates a regime belief.
- How the fit locates a break in time.
- How the soft answer smooths the trade.

Soft Regimes and Conviction Sizing

One bad day should not flip you fully

A strategy works in calm markets and should stand down in stressed ones. A hard label snaps between full size and zero on a single threshold. Instead we compute the probability of stress from today's return, and let it scale the position smoothly.

The setup

- Two regimes: calm and stressed.
- The trade is favorable only in calm.
- Today brings one sharp return.

What we'll do by hand

- Update $P(\text{stress})$ with one Bayes step.
- Size the trade by conviction.
- Contrast it with a hard label.

The Regime Posterior

Bayes, applied to today's return

$$P(\text{stress} \mid r) = \frac{P(r|\text{stress}) P(\text{stress})}{P(r|\text{stress}) P(\text{stress}) + P(r|\text{calm}) P(\text{calm})}$$

Reading it

- Weigh each regime's prior by how well it explains today.
- A big move is far more likely under the stressed regime.
- Normalize, and you get $P(\text{stress})$ given the return.

Why it's soft

- The answer is a probability, not a label.
- It moves smoothly as evidence arrives.
- It carries its own confidence.

The Setup: Two Regimes

Priors, volatilities, and today's return

	Calm	Stressed
prior probability	0.80	0.20
daily volatility	1%	3%

Today's return is -3%. Goal: compute $P(\text{stress} \mid -3\%)$ with one Bayes step on the next slide.

P(stress) by Bayes

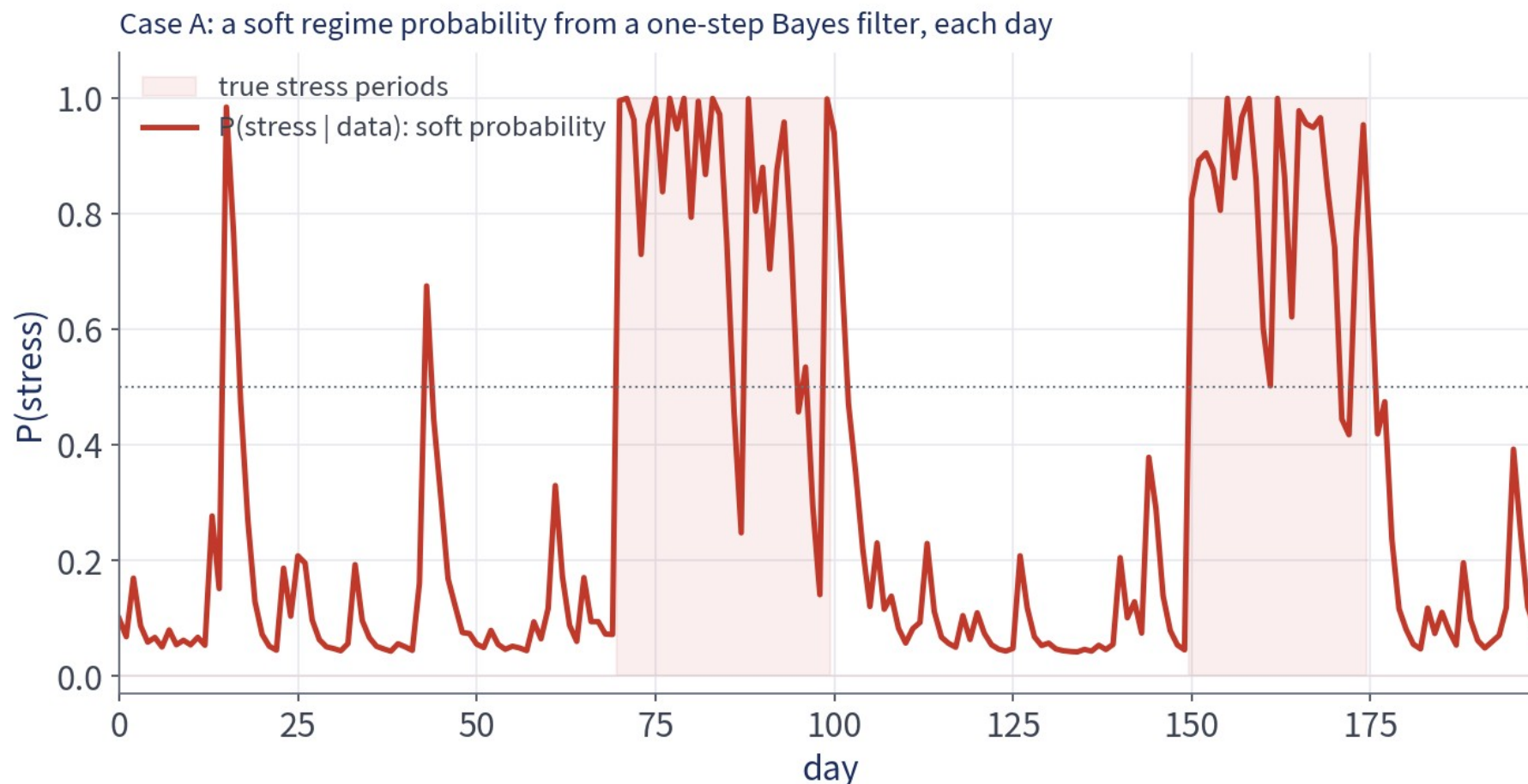
One update, three lines

$$\begin{aligned} \text{prior: } P(\text{calm}) &= 0.80, \quad P(\text{stress}) = 0.20; \quad \sigma_{\text{calm}} = 1\%, \quad \sigma_{\text{stress}} = 3\% \\ \text{today } r = -3\% : \quad L_{\text{calm}} &= \frac{1}{1}e^{-9/2} = 0.011, \quad L_{\text{stress}} = \frac{1}{3}e^{-9/18} = 0.202 \\ P(\text{stress} \mid r) &= \frac{0.202(0.20)}{0.202(0.20) + 0.011(0.80)} = \frac{0.0404}{0.0493} = \mathbf{0.82} \end{aligned}$$

A single -3% day lifts the stress probability from 0.20 to 0.82 - a strong but still soft signal, not a hard flip.

The Soft Regime Probability

Repeating that Bayes step every day gives a smooth $P(\text{stress})$ that rises into the stressed periods and falls out of them. It is a probability with shades of grey, never a jumpy 0/1 label.



Size by Conviction

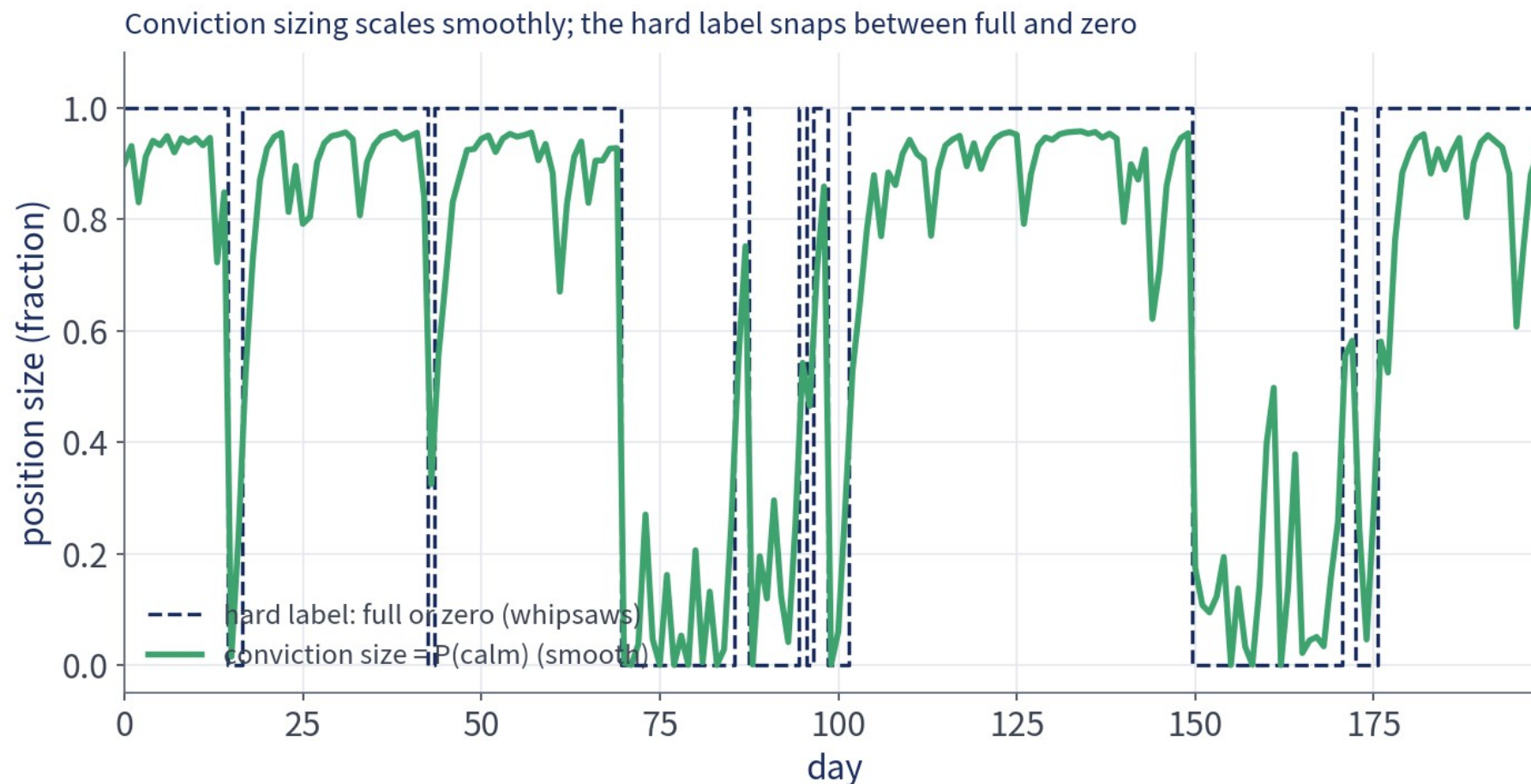
Soft probability vs hard label

$\text{size} = \text{edge} \times P(\text{favorable}), \quad \text{favorable} = \text{calm}$
before the -3% day: soft = 0.80, hard label = FULL
after ($P(\text{calm}) = 0.18$): soft = **0.18**, hard label = **ZERO**

The soft rule trims the position from 0.80 to 0.18 as stress rises. The hard label instead slams from full to zero on one day.

Conviction Sizing vs a Hard Label

The conviction size, equal to $P(\text{calm})$, glides up and down with the evidence. The hard label snaps between full and zero at the threshold - whipsaws in and out on borderline days and paying the costs each time.



What the Hard Label Costs

Whipsaw, turnover, and false certainty

With the hard label

- One threshold decides full or zero.
- Borderline days flip you in and out.
- Turnover and slippage pile up.

With conviction sizing

- The size tracks the probability smoothly.
- Borderline days get a middling weight.
- You trade less on genuine ambiguity.

When Did the Relationship Break?

A one-time shift at an unknown date

A trading relationship - a spread level, a strategy's mean - shifts once and stays shifted. We find WHEN by trying each possible break date and seeing which split fits best, then act on the post-break regime, sized by how sure we are of the date.

The setup

- A short series that jumps to a new level.
- The break date is unknown.
- Fitting one mean to all of it fits badly.

What we'll do by hand

- Compare no-break vs break fits.
- Score each candidate break date.
- Turn the scores into a posterior.

The Data: Six Months

Find where the level shifts

Month	1	2	3	4	5	6
Value (%)	+0.5	-0.2	+0.9	+1.7	+2.4	+2.0

Goal: does one mean fit, or did the level jump? We compare sums of squared errors (SSE) for each split.

No Break vs a Break

Compare how well each fits

no break: one mean 1.22, $SSE = 4.87$
break after month 3: $m_1 = 0.40$, $m_2 = 2.03$, $SSE = 0.87$
the break model fits about $5.6 \times$ better $\Rightarrow$ a break is real

Splitting after month 3 cuts the SSE from 4.87 to 0.87 - a far better fit. A break is real; now, exactly when?

Which Date? A Posterior

Score every candidate break

SSE if break after month 2, 3, 4 : 1.45, **0.87**, 1.97

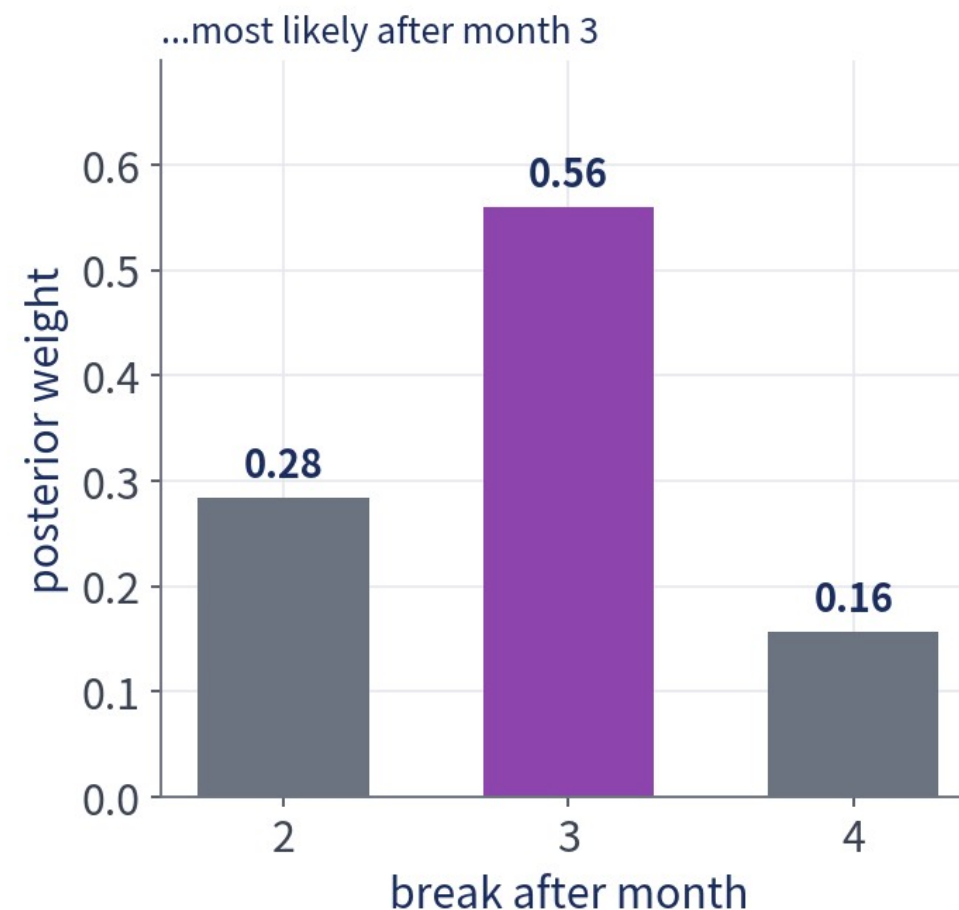
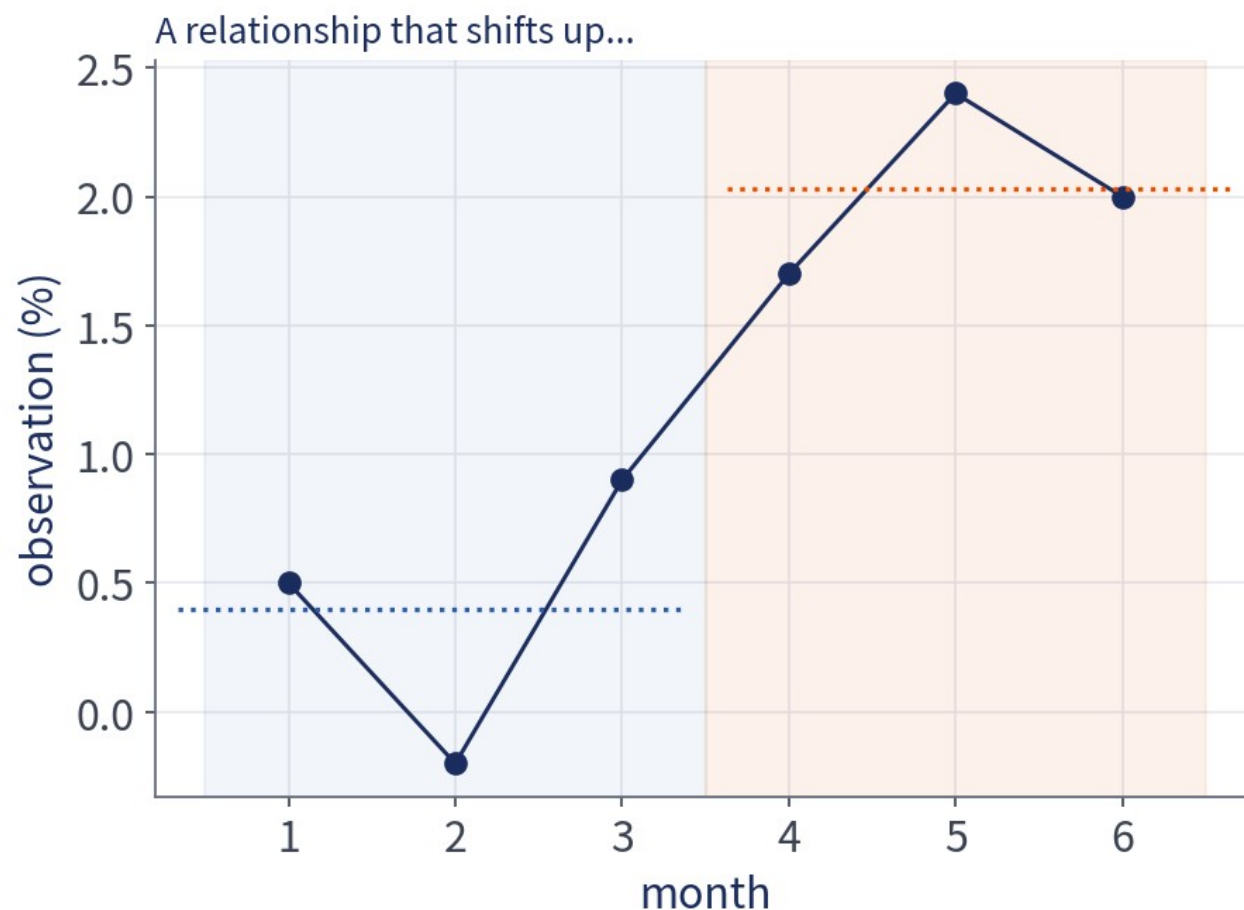
turn into weights $\propto e^{-\text{SSE}/2s^2}$: 0.28, **0.56**, 0.16

most likely after month 3 — but month 2 is plausible too

The best split is after month 3 (weight 0.56), but month 2 keeps a real 0.28. The break date itself is uncertain - and honestly so.

A Break With an Uncertain Date

The series shifts up around month 3 (left). Scoring each split turns into a posterior over the break date (right): it peaks after month 3 but spreads to month 2, telling us roughly - not exactly - when the rules changed



Act on the Break

Re-estimate, and size by certainty

Re-estimate after the break

- Use only the post-break data for the new regime.
- Old data would drag the estimate toward the past.
- The mean is now 2.03, not the blended 1.22.

Size by certainty of the date

- A sharp break date means act with confidence.
- A fuzzy one means blend across possible dates.
- Uncertainty about WHEN flows into the trade.

What Ignoring the Break Costs

Trading a stale relationship

If you ignore it

- You keep using the old, blended average.
- Your signal is centered on a level that's gone.
- Every trade is biased toward the past regime.

If you detect it

- You switch to the post-break parameters.
- Your signal reflects the current regime.
- You wait for more data when the date is unclear.

The Label-Switching Trap

Before you average regime estimates

What can go wrong

- When you fit regimes, their names are arbitrary.
- The sampler can swap 'calm' and 'stressed' mid-run.
- Averaging the raw draws then blends the two.

The simple fix

- Impose an order, e.g. calm has the lower volatility.
- That pins the labels down.
- Always check before trusting a regime average.

Conviction Sizing, in One Line

Both cases feed this

$$\text{position}_t \propto \underbrace{\text{edge}_t}_{\text{signal}} \times \underbrace{P(\text{favorable} \mid y)}_{\text{conviction}} \times \underbrace{\frac{1}{\text{band width}}}_{\text{less doubt, more size}}$$

The three inputs

- The edge: the raw strength of the signal.
- The conviction: $P(\text{favorable regime})$, from Case A.
- The certainty: tighter when the break date is clear.

The behavior

- Full size only when all three agree.
- Automatic caution when any is shaky.
- Uncertainty sets the bet, not a footnote.

The Common Pattern

Two calculations, one shape

Both cases, computed by hand, turn a soft belief into a sized action. A Bayes step gives a regime probability; a fit comparison gives a break date and its uncertainty. In each, that uncertainty flows straight into how much you trade.

The trap

- A hard yes/no on a regime or a break.
- Full commitment on a borderline call.
- Whipsaw and bias when the call is wrong.

The fix

- Compute a probability, not a label.
- Locate a break with its own uncertainty.
- Size the trade to that honest confidence.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full series.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - soft regime:

- * from priors 0.8/0.2 and vols 1%/3%, do one Bayes step on $r = -3\%$; print $P(\text{stress})$;
- * run the filter over a series and size by $P(\text{calm})$;
- * compare to a hard-label rule (turnover, whipsaw).

CASE B - break:

- * take the 6-month series; compute SSE for no break and for a break after months 2, 3, 4;
- * turn the SSEs into break-date weights;
- * re-estimate the post-break mean.

Match every printed number to the slides.

Check against the slides

- $P(\text{stress} \mid -3\%) = 0.82$.
- Soft size 0.80 $\rightarrow$ 0.18; hard full $\rightarrow$ zero.
- Break SSE 4.87 $\rightarrow$ 0.87 (after month 3).
- Date weights 0.28 / 0.56 / 0.16.

Deliverable: `part3_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does one -3% day raise $P(\text{stress})$ to 0.82?

**Bayes: weigh priors
by each regime's fit**

Q2 How do you locate the break date?

**The split with the
lowest SSE**

Q3 Why size by $P(\text{favorable})$ instead of a hard label?

**It scales smoothly
and avoids whipsaw**

All three come from the Bayes step, the SSE comparison, and conviction sizing.

Three Pitfalls to Avoid

Where these applications slip

Common mistakes

- "A hard regime label is decisive, so it's better."
- "The break date is known once I see the jump."
- "I can average raw regime draws directly."

The reality

- A soft probability avoids whipsaw and states its doubt.
- The date has a posterior - month 2 was plausible too.
- Fix label switching first, or the regimes blur.

Two Calculations to Remember

Part 3 applications in one screen

1

One Bayes step turns a return into a regime probability

Priors 0.80/0.20 and vols 1%/3% take a -3% day to $P(\text{stress}) = 0.82$. Sizing by $P(\text{calm})$ trims the trade from 0.80 to 0.18 smoothly, where a hard label would slam from full to zero.

2

Comparing fits locates a break - and its uncertainty

Splitting after month 3 cut the SSE from 4.87 to 0.87. The break-date weights (0.28, 0.56, 0.16) say 'most likely month 3, but month 2 is plausible' - honest uncertainty about when.

3

Soft beliefs become sized actions

Conviction sizing multiplies edge by the regime probability and by certainty about the break. Uncertainty is not a footnote; it sets the bet - the heart of trading with a posterior.

Next - Week 16: Synthesis and the Final Project, where the whole course comes together.